

Referential Regimes: Identity and Persistence Structure for Neutral Substrates

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Abstract

Modern data systems increasingly operate under conditions of persistent legal, political, and analytic disagreement. In systems that adopt a neutral substrate, shared interpretation cannot be assumed, and representations must preserve stable reference across incompatible extensions. This paper examines the structural consequences of that architectural choice. Treating interpretive neutrality and stable reference as explicit design constraints, the analysis asks what structure is required for entities and their relations to remain referable and comparable under persistent disagreement. Minimality is defined structurally: reductions are admissible only if they do not reintroduce identity-relevant distinctions through hidden roles, flags, or contextual predicates. A current lower bound is established: under the stated assumptions, a neutral substrate operating under persistent disagreement must realize at least nine distinct referential regimes. The result is a lower-bound result, not an upper-bound or completeness claim.

Preview of Main Result. *Under explicit neutrality and stability constraints, attempts to simplify structure by collapsing distinctions reintroduce*

those distinctions as hidden interpretive mechanisms. Excluding such hidden regimes implies nine distinct referential regimes required for accountability under persistent disagreement: fewer do not satisfy the stated requirements.

Informally. Given a neutral substrate capable of supporting persistent interpretive disagreement, entities and relations must remain stably referable across incompatible interpretive frameworks. This requirement constrains how entities may be individuated. Attempts to simplify the individuation structure by collapsing categories reintroduce those distinctions as hidden interpretive mechanisms.

Keywords: formal ontology; identity and persistence; neutral substrates; extension stability; neutral substrates; accountability; causal commitment; normative commitment;

1 Introduction: Structure Under Disagreement

Modern data systems increasingly operate across settings characterized by persistent legal, political, and analytic disagreement, where no single interpretive authority can be assumed. Three recurring settings make the structural requirements concrete. First, in multi-jurisdiction regulatory reporting, a single corporate entity operates under overlapping legal regimes that individuate obligations, violations, and responsible parties differently; audit trails must remain comparable across jurisdictions without presupposing which regime governs. Second, in legislative history tracking, statutory authority is amended, delegated, and contested over time, requiring stable reference to normative structures independently of the occurrences that enact or challenge them. Third, in cross-institutional provenance analysis, research or administrative records are produced, transferred, and re-analyzed across institutions whose classificatory and causal assumptions diverge, yet must remain referable as the same records across those divergent frameworks. These cases are not offered as empirical evidence; they function as structural diagnostics that make concrete the requirements of Section 4.

However, the design objective considered in this paper is different. The goal is not representational adequacy within a shared interpretive framework, but substrate stability across incompatible interpretive contexts. This shifts the design problem from domain modeling to the individuation of entities within a neutral substrate. When ontologies are deployed as shared substrates under persistent disagreement without additional constraints, commitments such as what counts as an agent, what individuates an event, and what constitutes an authority, may become sites of interpretive conflict rather than stable reference points.

No single interpretive authority can be presupposed, and semantic convergence cannot be enforced without excluding legitimate participants or perspectives. Nevertheless, shared use of data remains necessary: entities must be referable, actions traceable, and records comparable across frameworks that may diverge in assumptions, methods, or evaluative standards. Recent work is consistent with the view that systems without explicit structural commitments further motivating the need for substrates that enforce stable reference structurally rather than relying on interpretive consistency.

A natural response might be to reduce ontological commitment by collapsing distinctions and representing entities, occurrences, authorities, and records as a single undifferentiated type, differentiated only by role predicates (e.g., *isAgent*, *isOccurrence*, *isAuthority*) or using contextual flags supplied at query or extension time. This achieves surface parsimony but not structural neutrality. Such encodings relocate essential distinctions from the substrate to interpretive mechanisms that are not represented at the substrate level, and whose consistent application requires precisely the shared commitments that persistent disagreement denies. We refer to these as *hidden regimes*: structural discriminations that are operationally necessary but ontologically unacknowledged. Such encodings make referential stability contingent on interpretive consensus that cannot be assumed under persistent disagreement, and are therefore incompatible with the optimization criterion adopted here.

This analysis identifies a design objective that is distinct from the gen-

eral problem of ontology construction. treating ontological neutrality as a structural design constraint. Neutrality is understood as interpretive non-commitment at the substrate level combined with stability under incompatible extension: a substrate must permit independent extensions to introduce causal or normative structure without requiring revision of the underlying representation. Explanatory and evaluative structure is externalized; the substrate is constrained to remain neutral by design.

The substrate is not intended to adjudicate causal or normative claims. Its role is limited to preserving referential comparability across independently evolving extensions. All causal explanation, normative evaluation, and domain-specific interpretation are delegated to extension layers.

Central to this constraint is the notion of stable reference, which in turn constrains how entities may be individuated within the substrate. Stable reference in this work concerns regime-level persistence conditions, not immunity to reinterpretation at extension layers. The precise assumptions defining persistent disagreement and extension without coordination are stated in Section 3. Given stable reference as a structural constraint, minimality cannot be defined simply as a reduction in primitive count.

Apparent simplifications may achieve surface parsimony by relocating essential distinctions into roles, flags, contextual typing, or other interpretive mechanisms. Such encodings do not eliminate structure; they merely relocate it to extension layers that presuppose shared interpretive commitments. Under persistent disagreement, those commitments cannot be assumed. Minimality therefore requires that all operationally necessary referential distinctions be explicit and structurally enforced at the substrate level.

Accountability is the specific capacity discussed in this paper. Accountability analysis requires that a neutral substrate maintain stable reference conditions across all entities it represents (R1), and support representation of obligation-bearing entities, normative structures, time-indexed occurrences, applicability contexts, and descriptive records (R2–R6), without embedding causal claims or normative judgments in the substrate itself. These are

introduced as representational capacities whose simultaneous support places structural constraints on any neutral substrate. Whether they can be realized within fewer referential regimes is addressed in Section 7; the question is whether a neutral substrate can support them at all, and if so, with what irreducible structural commitments. The diagnostic cases introduced above (multi-jurisdiction regulatory reporting, legislative history tracking, and cross-institutional provenance) recur throughout the paper as structural illustrations.

The central contribution of this paper is a constraint result: a conditional lower bound on ontological structure under the design objectives stated above. Under explicit constraints of neutrality, stability under incompatible extension, and exclusion of hidden structural discriminations, we show that any substrate capable of supporting accountability under persistent disagreement must realize at least nine distinct referential regimes. The result is a constraint on what is structurally possible under a specific optimization objective, not a proposal for a universal ontology. Alternative objectives, such as interpretive convergence, domain-specific efficiency, or context-dependent substrate classification, yield different tradeoffs and are not addressed here.

Systems of practical and theoretical interest frequently exhibit persistent interpretive disagreement. In such settings, competing frameworks assign different interpretations to the same underlying structure, while the structure itself remains unchanged.

A representation suitable for such systems must therefore satisfy a neutrality requirement: it must support stable reference and structural comparison without committing to any particular interpretive framework.

establishes constraints for such neutrality, requiring that the substrate exclude causal and normative commitment while preserving extension-invariant identity.

The present paper addresses the representational problem: whether such systems admit a structural representation that is both expressive and neutral.

We proceed by restricting attention to admissible relations and defining a path-based grammar for multi-stage structure. We show that all admissible finite systems can be realized as families of paths over a minimal relation set. We then construct a regime-typed directed multigraph embedding that preserves entity identity, regime typing, and admissible relations, and is faithful up to regime-typed isomorphism.

Within this representation:

- all admissible structure is realized as path families,
- structural comparison reduces to comparison of such families,
- interventions correspond to admissible transformations of path structure.

Crucially, the representation introduces no causal or normative predicates. Such interpretations arise only in extensions over REC-anchored paths.

This establishes that admissible systems under persistent disagreement can be represented, compared, and transformed entirely at the level of neutral structure.

The remainder of the paper proceeds as follows. Section 2 situates the analysis as a refinement of the neutral-substrate constraint introduced in SE-100. Section 3 states the assumptions, neutrality constraints, and optimization criterion. Section 4 formalizes the structural requirements imposed by accountability analysis. Section 5 fixes the formal vocabulary used in the derivations. Section 6 introduces the formal setting, including admissible transformations, regime profiles, and path-grammar adequacy.

Section 7 derives the six-family lower bound: the coarse requirements force distinct referential regime families before transformation-driven refinement is considered. Section 7 gives the corresponding six-family construction before refinement. Section 9 then derives the required refinements. In particular, Sections 9.6, 9.7, and 9.8 show that enduring non-normative referents, applicability contexts, and normative structures split into distinct

referential regimes under transformation pressure.

Section 10 states the resulting nine-regime lower bound. Section 11 constructs the neutral regime embedding. Section 12 derives consequences for interpretation, comparison, and intervention. Section 13 discusses boundary conditions and the scope of the result. Section 14 situates the work relative to existing frameworks. Section 15 states limitations, including the absence of an upper-bound or completeness claim. Section 16 concludes.

2 Background: Neutral Substrates

This paper refines the referential-regime component of the neutral-substrate constraint introduced in SE-100. In SE-100, a neutral substrate is a shared foundational layer whose neutrality is guaranteed at design time by restricting substrate-layer commitments to referential commitments and permitted attribution propositions. It makes no object-level substrate-layer commitments to causal or normative propositions.

The present paper analyzes the identity and persistence structure required for those referential commitments to remain usable as common ground under persistent interpretive disagreement. The regimes derived below should therefore be read as refinements of the referential regime required by SE-100, not as a general-purpose upper ontology.

3 Assumptions and Design Constraints

This section fixes the problem setting for the remainder of the paper. All results that follow are conditional on the assumptions stated here. The purpose is not to argue that these assumptions are universally correct, but to make explicit the constraints under which a neutral substrate for accountability is possible. These constraints determine how entities may be individuated within such a substrate.

3.1 Persistent Disagreement and Extension Without Coordination

We assume environments in which disagreement is not an anomaly to be resolved, but a persistent condition. Legal interpretation, policy evaluation, and analytic methodology may remain incompatible across communities and over time. Accordingly, no single interpretive authority or shared semantic framework is assumed at the substrate level.

We further assume that extensions to the substrate may occur without coordination. Independent actors may add explanatory, normative, or analytic structure, and such extensions must not require revision of the substrate itself. This assumption does not rely on global semantic agreement, centralized governance, or negotiated meaning as preconditions for interoperability. Under these assumptions, substrate-level admissibility must not depend on interpretive agreement among extensions.

3.2 Stability Under Persistent Disagreement

Stable reference in this work concerns regime-level identity and persistence conditions, not immunity to transformation or versioning at interpretive layers. The substrate must preserve the referential integrity of entities and records even as their interpretation evolves in incompatible ways across extensions. Persistent disagreement, as used in this work, is defined by four conditions:

- Extensions may introduce incompatible causal accounts.
- Extensions may introduce incompatible normative attributions.
- Extensions cannot be forced into convergence.
- Substrate revision is not permitted to resolve interpretive conflict.

Under these conditions, the substrate must remain interpretively agnostic while preserving referential stability across extensions. Stability is a property

of the referential regime that governs individuation, rather than any particular token or interpretive assignment within it.

3.3 Neutrality Constraint

This work assumes substrate neutrality and stability as a design constraint rather than as a philosophical claim. A neutral substrate does not assert why events occur, whether outcomes are desirable, or how actions ought to be evaluated. Instead, it provides stable referents and structural relations upon which external interpretive layers may operate.

Neutrality further requires stability under incompatible extension at the level of identity conditions and admissible relations. If two extensions disagree in interpretation, evaluation, or explanation, that disagreement must not force reclassification of entities, alteration of identity conditions, or revision of the substrate. Any representation whose admissibility conditions depend on interpretive agreement does not satisfy this constraint.

3.4 Stable Reference Requirement

For a substrate to support accountability, entities must remain referable over time and across extensions. Identity conditions and admissible relations must not depend on interpretive convergence. If an entity must change its ontological status in order to participate in new relationships or analyses, stable reference is not preserved. This requirement is not satisfied by representations in which identity is determined contextually or retroactively by interpretive layers. The domain consequences of this requirement are illustrated in Section 4 under R1.

This requirement excludes representations in which stability-critical distinctions are encoded only as roles, flags, or contextual predicates. Such mechanisms rely on interpretive conventions supplied by extension layers and therefore permit divergent admissibility judgments under persistent

disagreement.

The stability requirement is intentionally strong. Engineering systems may address identity pressure through contextual or negotiated identity schemes, and in such settings strong reference stability may not be attainable in practice. The present analysis instead examines the case where such mechanisms are unavailable and stable reference must be preserved independently of interpretive convergence.

3.5 Optimization Criterion

The optimization objective of this paper is to minimize explicit ontological structure subject to the constraints above. Minimality is not defined as the smallest possible number of primitives in the abstract, but as the smallest number of explicit referential regimes required to preserve neutrality and stable reference.

A proposed reduction constitutes a genuine simplification only if it does not reintroduce the collapsed distinctions through hidden discriminators, interpretive mechanisms, or role predicates.

The minimality criterion adopted here is analytical rather than prescriptive: it is introduced to determine what ontological structure is unavoidable when neutrality and stability under persistent disagreement are treated as design constraints.

Alternative optimization objectives prioritize different tradeoffs, such as interpretive coordination, domain-specific efficiency, or context-dependent classification at the substrate level. The results presented here apply only to substrates optimized for interoperability under persistent disagreement.

These assumptions and constraints determine the design space within which a neutral accountability substrate must operate. The following section states the representational capacities that any substrate must satisfy under these conditions.

4 Structural Requirements for Neutral Substrates

Given the assumptions and optimization criterion fixed in Section 3, the central question is not which ontology is most expressive or convenient, but which structural distinctions are unavoidable. We therefore state the representational requirements that an ontological substrate must satisfy in order to support accountability analysis under persistent disagreement.

These requirements are structural rather than interpretive. They do not prescribe how accountability should be understood, evaluated, or enforced; instead, they specify what must be representable in a stable and neutral manner.

The requirements are framed as necessary capacities. They are not derived from any particular domain, policy area, or theory of governance, but from the minimal conditions under which accountability claims can be stated, inspected, and compared without embedding causal or normative commitments in the substrate.

Before deriving the lower bound, we make explicit what is meant by a *referential regime*.

Definition 4.1 (Referential Regime). An referential regime specifies the conditions under which entities remain the same entities over time and across incompatible extensions, together with the relations they may admit without altering their identity conditions. Referential regimes therefore represent distinct modes of individuation within a neutral substrate.

Two entities belong to the same regime if and only if they share the same stability-critical identity criteria, persistence conditions, and admissible relations under the neutrality and stability constraints adopted here.

Regimes are not introduced as ontological kinds or domain categories. They are structural distinctions forced by the requirement that identity and admissibility remain invariant under persistent disagreement.

The requirements enumerated below differ in scope. R1 is a cross-cutting constraint on individuation: it applies to every entity representable in the substrate, regardless of which other requirements govern its regime. R2–R6 each specify representational capacities not entailed by R1 alone.

4.1 R1: Stable Identity and Persistence (Global Constraint)

Requirement. The substrate must support stable reference to all entities it represents over time.

R1 operationalizes the Stable Reference Requirement (Section 3.4) as a representational capacity governing how entities are individuated and tracked within the substrate. Changes in interpretation, role, status, or evaluation must not require reclassification of entities or modification of their identity criteria. Representations that determine identity contextually or retroactively through interpretive layers fail to satisfy R1: their identity conditions are not invariant under extension and therefore do not constitute stable reference for accountability purposes. In addition, admissible relations must be invariant under extension: whether a relation is well-formed between entities of given regimes must not depend on interpretive framework or extension-layer agreement. No relation may determine or alter the identity conditions of the entities it connects, nor introduce or encode identity criteria through participation in relations.

In cross-institutional provenance, a research record transferred between institutions must remain referable as the same record regardless of how each institution classifies, annotates, or re-analyzes it. If identity conditions shift when an institution introduces a new classificatory scheme, the provenance chain breaks and accountability for the record’s content and custody cannot be maintained. R1 is therefore non-optional wherever records must persist as stable referents across frameworks that may individuate them differently.

4.2 R2: Obligation-Bearing Capacity

Requirement. The substrate must support representation of entities that can bear obligations or responsibilities independently of particular occurrences.

Such entities must be referable independently of the particular occurrences in which obligations are fulfilled, violated, or discharged. This requirement does not presuppose any substantive theory of normativity. It requires only that obligation-bearing entities be representable as enduring referents whose identity does not depend on specific events.

In multi-jurisdiction regulatory reporting, a corporate entity must be referable as an obligation-bearer across legal regimes that may differ in how they define corporate personhood, subsidiary responsibility, or delegated liability. If the substrate cannot represent obligation-bearing entities independently of the specific events in which obligations are fulfilled or violated, cross-jurisdiction audit comparisons require interpretive agreement on the nature of the bearer itself, precisely the agreement that cannot be assumed under persistent disagreement. R2 is therefore non-optional wherever obligation-bearers must be referable across legal or institutional frameworks that individuate corporate personhood, subsidiary responsibility, or delegated liability differently.

4.3 R3: Normative Reference Without Execution

Requirement. The substrate must distinguish between the existence of authority or obligation and the occurrences that execute, comply with, or violate it.

Normative or regulatory structures must therefore be representable independently of the occurrences that enact, comply with, or violate them. This distinction is necessary to support accountability over time, including cases of delayed enforcement, partial compliance, amendment, or contested application. Encoding normative reference solely through occurrences col-

lapses authority into execution and undermines stable reference to governing structures.

In legislative history tracking, a statutory authority must be referable as a normative structure independently of the legislative acts that enacted it, the administrative actions that applied it, and the judicial decisions that interpreted it. If normative reference is encoded solely through occurrences, a statute that has been amended but not yet enforced, or contested but not yet resolved, has no stable substrate-level referent. R3 is therefore non-optional wherever accountability requires distinguishing compliance with authority-as-written from compliance with authority-as-executed, including cases of amendment, delayed enforcement, or contested application.

4.4 R4: Time-Indexed Occurrence

Requirement. The substrate must support representation of occurrences individuated by temporal realization and provenance.

Occurrences must be individuated by temporal realization and provenance, and must remain distinct from enduring entities. This requirement applies to all discrete occurrences relevant to accountability, including actions, filings, inspections, or transactions. The list is illustrative; the requirement extends to any occurrence for which temporal individuation and provenance are necessary for stable reference. Occurrences are not required to carry interpretive or evaluative meaning; they must assert only that something occurred at a particular time and place.

In multi-jurisdiction regulatory reporting, a filing, inspection, or transaction must be individuated by when and where it occurred, independently of how different jurisdictions classify or evaluate it. Two jurisdictions may disagree on whether a given action constitutes compliance, but both must be able to refer to the same occurrence as their object of evaluation. Without substrate-level individuation of time-indexed occurrences, that shared reference is unavailable and cross-jurisdiction comparison reduces to a dispute about

what happened rather than about how to evaluate it. R4 is therefore non-optional wherever cross-jurisdiction accountability requires shared reference to occurrences independently of how those occurrences are classified or evaluated.

Requirements R2, R3, and R4 are mutually irreducible as representational capacities. R2 requires stable reference to an enduring bearer; R3 requires stable reference to a governing normative structure; R4 requires stable reference to a discrete, time-indexed occurrence. Satisfying any one does not entail satisfying the others. None of the three capacities is definable in terms of the remaining two without either circularity or the collapse of distinctions that R1 requires to remain stable across incompatible extensions.

4.5 R5: Applicability and Scope

Requirement. The substrate must support representation of applicability contexts that scope where, to whom, or under what conditions a normative structure applies.

It must be possible to state where, to whom, or under what conditions a normative or regulatory structure applies, without collapsing such contexts into either obligation-bearing entities or physical loci. Applicability contexts may be nested, overlapping, or partially coincident. They must therefore be representable as stable referents in their own right, rather than as incidental attributes whose interpretation may vary across extensions.

Applicability contexts cannot be reduced to properties of the R3 normative structures they scope: a single normative structure may admit different applicability contexts under different jurisdictional frameworks, and collapsing scope into the structure would require its identity conditions to vary with those frameworks, violating R1. Nor can applicability contexts be reduced to obligation-bearing entities (R2), since scope specifies the domain over which authority applies rather than identifying any particular bearer.

In multi-jurisdiction regulatory reporting, normative structures apply to different sets of entities depending on jurisdiction, and those jurisdictions may be nested, overlapping, or contested. If applicability contexts cannot be represented as stable referents in their own right, the scope of an obligation can only be recovered by interpreting the obligation itself which reintroduces the shared interpretive assumptions the substrate is designed to avoid. R5 is non-optional wherever accountability claims must be scoped without presupposing agreement on which normative framework or authority governs.

R2–R5 together establish the referential structure required for accountability analysis: obligation-bearing entities (R2), governing normative structures (R3), time-indexed occurrences (R4), and the applicability contexts that scope those structures (R5). These capacities make it possible to represent who is responsible, which authority governs, what occurred, and where that authority applies. They do not, however, establish stable reference to the descriptive records or indicators associated with those entities and occurrences.

4.6 R6: Descriptive Indicators Without Causal Commitment

Requirement. The substrate must support representation of descriptive indicators or records that characterize states, performance, or outcomes associated with entities or occurrences without asserting causal or normative claims.

R6 is not entailed by R1–R5. Stable reference to obligation-bearing entities (R2), normative structures (R3), time-indexed occurrences (R4), and applicability contexts (R5) does not establish stable reference to descriptive records associated with those referents. Descriptive indicators have individuation conditions that are independent of temporal realization (R4), do not assert or constitute normative authority (R3), do not identify obligation-bearers (R2), and do not scope the application of normative structures (R5). They must therefore be specified by a distinct requirement.

Such indicators must be representable without asserting causal relations. This requirement does not preclude the representation of causal claims at extension layers. This requirement is necessary to support longitudinal comparison and outcome inspection while preserving neutrality. Collapsing indicators into occurrences or into interpretive annotations eliminates stable reference to descriptive records over time and across analytic frameworks.

In cross-institutional provenance, performance indicators and outcome records must remain comparable across institutions whose causal and normative frameworks differ, for instance, where one institution models outcomes in terms of intervention efficacy and another in terms of systemic risk factors. If descriptive indicators are encoded as occurrences, they carry implicit individuation conditions tied to causal assumptions; if encoded as interpretive annotations, they are unstable under extension. R6 is non-optional wherever longitudinal comparison of records must be preserved without committing the substrate to any particular explanatory or evaluative framework.

4.7 Summary

Together, R1–R6 specify the minimal representational capacities required for an accountability substrate operating under persistent disagreement. R1 is a cross-cutting stability constraint that applies to all entities the substrate must represent. R2–R5 specify four distinct representational capacities that are mutually irreducible: no one of them is entailed by or definable in terms of the others. R6 specifies a further capacity not entailed by R1–R5. These requirements do not impose a particular ontological partition; they define the structural constraints under which any such partition must operate. The following section uses R1–R6 to derive a lower bound on the number of distinct referential regimes a substrate must realize in order to satisfy them simultaneously.

5 Preliminaries

This paper refines the referential-regime component of the neutral-substrate constraint introduced in SE-100. In SE-100, a referential regime consists of individuation, co-reference, and persistence conditions. The present paper analyzes the persistence and transformation behavior needed for those referential commitments to remain usable as common ground under persistent interpretive disagreement.

This section fixes the formal vocabulary used in the derivations. Identity is represented via regime-specific equivalence relations and the quotient structures induced by those relations. Statements are formulated in terms of equivalence classes; the derivations in this section rely on invariance under admissible transformations.

5.1 Neutral Substrate Constraint

Definition 5.1 (Neutral substrate). A neutral substrate is a representation layer that does not encode downstream interpretive content.

Definition 5.2 (Extension). An extension is any admissible addition, refinement, or reorganization of representations that preserves neutrality.

Definition 5.3 (Extension stability (extension invariance)). A condition is extension-stable (equivalently, extension-invariant) if it is preserved under all admissible extensions.

Definition 5.4 (Admissible ontology). An ontology is admissible if it has explicitly specified identity and persistence conditions that are extension-stable.

The neutral substrate represents entities without fixing downstream interpretation. Identity and persistence must be stable under extensions by heterogeneous and potentially conflicting interpretations.

5.2 Representation Space

This subsection fixes the representation domain on which all regimes and transformations are defined.

Definition 5.5 (Representation space). Let $\mathcal{R}$ denote the set of all well-formed representations that satisfy neutrality and stable reference.

5.3 Transformation Basis

This subsection fixes the transformation families that parameterize identity and persistence behavior.

Definition 5.6 (Transformation family). Let $\mathbb{F}$ be a fixed set of transformation families:

$$\mathbb{F} = \{\text{RE}, \text{AN}, \text{RF}, \text{AD}, \text{RC}, \text{RA}, \text{SU}, \text{BF}, \text{PV}, \text{SE}\}.$$

Each $f \in \mathbb{F}$ denotes a transformation family, i.e., a class of admissible extension-relative transformations on $\mathcal{R}$.

5.4 Regime Profiles

This subsection defines regime profiles as the structures that determine identity, applicability, classification, and persistence.

Definition 5.7 (Applicability). For a regime profile P , the applicability map

$$\text{Appl}_P : \mathbb{F} \rightarrow \{0, 1\}$$

is defined by

$$\text{Appl}_P(f) = 1$$

iff the transformation family f is applicable under P .

Definition 5.8 (Transformation classification). For a regime profile P , the classification map is the function

$$\text{Class}_P : \{f \in \mathbb{F} \mid \text{Appl}_P(f) = 1\} \rightarrow \{\text{IGN}, \text{PRS}, \text{BRK}\}.$$

Thus classification is defined only for applicable transformation families.

For transformation families f with $\text{Appl}_P(f) = 0$, classification is undefined. In profile tables, such entries are marked N/A, indicating inapplicability rather than an unclassified applicable transformation.

Definition 5.9 (Persistence condition). For a regime profile P , the persistence condition

$$\text{Persist}_P \subseteq \{f \in \mathbb{F} \mid \text{Appl}_P(f) = 1 \text{ and } \text{Class}_P(f) = \text{PRS}\}$$

specifies which applicable identity-preserving transformation families count as continuation of the same entity.

Identity is given by equivalence classes; persistence governs continuity under transformations classified as preserving identity.

Definition 5.10 (Regime profile). A regime profile is a tuple:

$$P = (C_P, B_P, \text{Appl}_P, \text{Class}_P, \text{Persist}_P)$$

where:

- C_P is the carrier of identity,
- B_P is the identity basis,
- $\text{Appl}_P : \mathbb{F} \rightarrow \{0, 1\}$ is the applicability map,
- Class_P is a function assigning each applicable transformation family a value in $\{\text{IGN}, \text{PRS}, \text{BRK}\}$,
- Persist_P specifies persistence behavior for transformations classified as preserving identity.

Definition 5.11 (Regime). A regime R implements a regime profile P if the identity basis, applicability map, transformation classification, and persistence behavior of R are exactly those specified by P , i.e., R realizes B_P , Appl_P , Class_P , and Persist_P on $\mathcal{R}$.

5.5 Admissible Transformations

This subsection defines admissibility conditions for transformations relative to a regime profile.

Definition 5.12 (Admissible transformation). A transformation $\tau : \mathcal{R} \rightarrow \mathcal{R}$ is admissible under a regime profile P if:

1. τ preserves well-formedness,
2. τ preserves neutrality,
3. τ preserves the invariants determined by the identity basis B_P , where B_P is the identity basis component of the regime profile P defined in Definition 5.10.

Definition 5.13 (Admissible transformation class). For a regime profile P , let T_P denote the set of all transformations admissible under P .

No closure or invertibility assumptions are imposed on T_P unless explicitly stated.

5.6 Identity and Quotients

This subsection defines identity as an equivalence relation induced by transformation classification and introduces the associated quotient structure.

Definition 5.14 (Induced identity). Let P be a regime profile. Define $\sim_P$ to be the equivalence relation on $\mathcal{R}$ generated by the transformation families

$f \in \mathbb{F}$ such that

$$\text{Appl}_P(f) = 1 \quad \text{and} \quad \text{Class}_P(f) = \text{PRS}.$$

Definition 5.15 (Equivalence class). The identity of x under P is its equivalence class:

$$[x]_P = \{y \in \mathcal{R} \mid x \sim_P y\}.$$

Definition 5.16 (Quotient structure). The set of entities under profile P is the quotient:

$$\mathcal{E}_P := \mathcal{R} / \sim_P.$$

All identity statements are interpreted at the level of equivalence classes in $\mathcal{E}_P$.

5.7 Invariance and Persistence

This subsection characterizes invariance conditions required for well-defined identity under a regime profile.

Definition 5.17 (Invariant). An invariant for a regime profile P is a property preserved under all applicable transformation families classified as PRS by Class_P .

Invariance under admissible transformations is a necessary condition for well-defined identity under a regime profile.

5.8 Profile Comparison

This subsection defines equivalence and non-collapse between regime profiles via their induced identity relations.

Definition 5.18 (Profile equivalence). Two regime profiles P, Q are equiva-

lent if they induce the same equivalence relation on $\mathcal{R}$, i.e.,

$$\sim_P = \sim_Q .$$

Equivalently, P and Q are equivalent if every identity class under P is also an identity class under Q , and vice versa.

If P and Q are equivalent, then $\mathcal{E}_P = \mathcal{E}_Q$.

Definition 5.19 (Collapse). Two regime profiles P, Q collapse if they are equivalent but not identical, i.e., $\sim_P = \sim_Q$ but $P \neq Q$.

Definition 5.20 (Non-collapse). Two regime profiles P, Q are non-collapsing if they are not equivalent, i.e., $\sim_P \neq \sim_Q$. Equivalently, P and Q are non-collapsing if there exists at least one pair of representations that are in the same identity class under one profile but not under the other.

5.9 Transformation-Driven Structure

This subsection formalizes transformation patterns and the conditions under which they are realized or force refinement of regime profiles.

Definition 5.21 (Coherent transformation pattern). A transformation pattern Π is a pair

$$\Pi = (\text{Appl}_\Pi, \text{Class}_\Pi)$$

where

$$\text{Appl}_\Pi : \mathbb{F} \rightarrow \{0, 1\}$$

and

$$\text{Class}_\Pi : \{f \in \mathbb{F} \mid \text{Appl}_\Pi(f) = 1\} \rightarrow \{\text{IGN}, \text{PRS}, \text{BRK}\}.$$

The pattern Π is coherent if:

- it satisfies the neutrality constraint,
- it induces a well-defined equivalence relation on $\mathcal{R}$,

- it contains no internal contradictions in applicability or classification.

Definition 5.22 (Profile realization). A regime profile P realizes a transformation pattern $\Pi = (\text{Appl}_\Pi, \text{Class}_\Pi)$ iff

$$\text{Appl}_\Pi = \text{Appl}_P \quad \text{and} \quad \text{Class}_\Pi = \text{Class}_P.$$

Definition 5.23 (Split pressure). A provisional regime profile P is under split pressure if there exists a transformation family $f \in \mathbb{F}$ such that no single pair

$$(\text{Appl}_P(f), \text{Class}_P(f))$$

is consistent with all admissible cases assigned to P .

Definition 5.24 (Forced regime profile). A regime profile is forced if there exists a coherent transformation pattern Π required for stable identity under extension such that no existing profile realizes Π .

Since Class_P is a function on applicable transformation families, each applicable $f \in \mathbb{F}$ receives exactly one classification under P . Accordingly, no profile assigns more than one classification to the same transformation family.

5.10 Minimal Regime Sets

This subsection defines minimal sets of regime profiles sufficient to realize all required transformation patterns without collapse.

Definition 5.25 (Minimal regime set). A set of regime profiles $\mathcal{P}$ is minimal if:

- every coherent transformation pattern required for stable identity under extension is realized by some $P \in \mathcal{P}$,
- no two profiles in $\mathcal{P}$ collapse,
- removing any profile from $\mathcal{P}$ violates one of the above conditions.

5.11 Obstruction and Separation

This subsection introduces obstruction and witness concepts used to establish non-collapse and necessity results.

Definition 5.26 (Obstruction). An obstruction is a structural condition that prevents a target property from being satisfied under the admissibility constraints.

Definition 5.27 (Witness). A witness is a concrete configuration demonstrating the presence of an obstruction.

Definition 5.28 (Irreducible obstruction). An obstruction is irreducible if no admissible transformation eliminates it.

Definition 5.29 (Separation witness). A separation witness for regime profiles P, Q is a pair $(x, y) \in \mathcal{R}^2$ such that $x \sim_P y$ and $x \not\sim_Q y$, or vice versa.

5.12 Logical Form of Results

This subsection specifies the form in which results are stated and proved.

Statements of results are formulated in terms of quotient structures $\mathcal{E}_P$. Proofs proceed by establishing invariance properties under admissible transformation classifications.

5.13 Representation Constraints

5.13.1 Admissibility Constraints

Proposition 5.30 (Admissibility Boundary). *Systems requiring primitive causal, normative, or evaluative relations are not representable under the admissible relation set.*

Proof. Primitive causal and normative relations are excluded by admissibility.

□

Proposition 5.31 (Causal Non-Representability). *The representation preserves temporal order but does not represent causal production.*

Proof. Only the admissible relation precedes is available, which encodes order but not causal production.

□

Proposition 5.32 (Normative Non-Representability). *Normative force is not represented; only structural placement of normative entities is preserved.*

Proof. No admissible relation encodes obligation, permission, or prohibition.

□

5.13.2 Representation Properties

Proposition 5.33 (REC Dependence). *Observability is mediated by REC-anchored paths.*

Proof. REC nodes provide the only linkage to descriptive records within the representation.

□

Proposition 5.34 (Granularity Dependence). *Path structure depends on modeling granularity.*

Proof. Refinement alters vertex and edge structure, thereby altering path families.

□

Corollary 5.35 (Path Sufficiency). *All admissible structure is representable as path families over admissible relations.*

Proof. Follows immediately from expressive adequacy of the path grammar; Section 6.

□

6 Formal Setting

This section introduces the structural objects over which regime profiles are defined. We first identify the types of structures that may carry identity, and then specify the transformation families under which identity must remain stable.

6.1 Identity Carriers

This subsection introduces the structural objects that may bear identity under regime profiles.

Definition 6.1 (Identity carrier). An identity carrier is a structural object in $\mathcal{R}$ whose identity is determined under a regime profile.

Identity carriers are structural positions within $\mathcal{R}$, not ontological types. Regime assignment depends solely on transformation behavior, not carrier category. Identity may be borne by different structural forms, each supporting distinct identity and persistence behavior under transformation.

Example 6.2 (Carrier types). Identity carriers include structural forms such as:

- enduring non-normative referents,
- normative structures,
- time-indexed occurrences,
- applicability contexts,
- descriptive records.

This list is illustrative; the analysis is not restricted to these forms.

These categories are not introduced as a fixed ontology. They serve to distinguish the structural forms over which identity conditions may be defined. Regime profiles refine these categories further based on transformation behavior.

6.2 Transformation Semantics

This subsection specifies the interpretation of the transformation families in $\mathbb{F}$.

Definition 6.3 (Transformation semantics). Each transformation family in $\mathbb{F}$ denotes a class of admissible operations on representations with the following structural interpretation:

- RE (re-expression): change of representation format preserving content,
- AN (annotation): addition of non-identity-bearing information,
- RF (refinement): increase in descriptive or structural detail,
- AD (aggregation/decomposition): restructuring into coarser or finer components,
- RC (re-contextualization): change in applicability context,
- RA (reassignment): change in association or bearer,
- SU (substitution): replacement of the identity carrier,
- BF (branch/fork): divergence into multiple continuations,
- PV (provenance extension): addition of historical or derivational information,
- SE (state evolution): change in state of an enduring referent over time.

These interpretations are structural rather than interpretive. They do not encode causal or normative commitments, and are intended to apply uniformly across domains. Each transformation family is defined extension-relatively and is evaluated only through its effect on equivalence relations induced by regime profiles.

Transformation families parameterize identity behavior. A regime profile assigns to each applicable transformation a classification determining whether identity is preserved, broken, or ignored. Distinct profiles therefore correspond to distinct invariance conditions over a fixed transformation basis.

6.3 Profiles as Transformation Constraints

This subsection characterizes how regime profiles constrain admissible relations via induced identity.

Definition 6.4 (Profile-governed relation). A relation $\rho \subseteq \mathcal{R} \times \mathcal{R}$ is profile-governed if its admissibility is determined solely by the equivalence classes $[x]_P$ of its relata under a fixed regime profile P .

Preservation of a profile-governed relation under transformation is equivalent to preservation of admissibility. Relations depending on role, institutional context, or higher-order grouping are not profile-governed and fall outside the scope of the present analysis.

A regime profile specifies how identity behaves under the transformation basis. Different profiles correspond to distinct invariance conditions over the same transformation families, inducing distinct equivalence relations on $\mathcal{R}$.

Profile-governed relations depend only on equivalence classes. Accordingly, all admissible structure is invariant under replacement of representatives within a class, and representation proceeds at the level of quotient structure.

The derivation of regime profiles proceeds by identifying coherent transformation-classification patterns required for stable reference, and separating those that cannot be realized by a single profile. When no single classification assignment for a transformation family is consistent across all admissible cases, refinement is required.

Extension-stable and extension-invariant are used synonymously.

6.4 Canonical Regime Classification Tables

This subsection records the transformation classification for the three regime profiles that do not exhibit split pressure under the transformation basis $\mathbb{F}$. Their classifications are stated here to make this paper self-contained and to

anchor the cross-family non-collapse arguments of Section 11.6.

6.4.1 OBL: Obligation-bearing entity

- RE: IGN
- AN: IGN
- RF: PRS
- AD: PRS
- RC: IGN
- RA: IGN
- SU: BRK
- BF: IGN
- PV: IGN
- SE: IGN

Identity is determined by persistence as a responsible party. Obligation-bearing entities do not undergo state evolution in the sense relevant to SE: their identity is not individuated by internal state transitions but by referential continuity as a bearer. Branching is identity-ignoring: the obligation-bearing entity persists independently of how its representations are forked.

6.4.2 OCC: Time-indexed occurrence

- RE: IGN
- AN: IGN
- RF: PRS
- AD: BRK (decomposition produces sub-occurrences with distinct temporal individuation; the original occurrence's identity is not preserved)
- RC: IGN

- RA: N/A ($\text{Appl}_{\text{OCC}}(\text{RA}) = 0$)
- SU: BRK
- BF: N/A ($\text{Appl}_{\text{OCC}}(\text{BF}) = 0$)
- PV: IGN
- SE: PRS

Identity is determined by temporal realization and provenance. State evolution is identity-preserving: an occurrence may have stages or phases that constitute the same event. Decomposition is identity-breaking: it produces sub-occurrences with distinct temporal individuation, and the original occurrence's identity is not preserved. Branching and reassignment are inapplicable because a time-indexed occurrence is fixed by its realization event and admits no forking or bearer reassignment while preserving that individuation.

6.4.3 REC: Descriptive record

- RE: IGN
- AN: PRS
- RF: PRS
- AD: PRS
- RC: IGN
- RA: IGN
- SU: BRK
- BF: PRS
- PV: PRS
- SE: IGN

Identity is determined by persistence as a descriptive referent without causal or normative commitment. Annotation, provenance extension, and branching are identity-preserving: a descriptive record persists across enrichment, derivation, and copying. State evolution is identity-ignoring: descriptive records do not have internal states in the sense relevant to SE.

Canonical classification summary.

	OBL	OCC	REC
RE	IGN	IGN	IGN
AN	IGN	IGN	PRS
RF	PRS	PRS	PRS
AD	PRS	BRK	PRS
RC	IGN	IGN	IGN
RA	IGN	N/A	IGN
SU	BRK	BRK	BRK
BF	IGN	N/A	PRS
PV	IGN	IGN	PRS
SE	IGN	PRS	IGN

7 Six-Family Lower Bound

The previous section specified the structural requirements for substrates under neutrality and stable reference. We now use those requirements to derive a lower bound on the individuation structure that any such substrate must realize.

The result is a necessity claim: if a substrate is to support accountability under persistent disagreement while maintaining neutrality and stable reference, some minimum number of distinct referential regimes is unavoidable. The lower bound is established by showing that certain representational capacities cannot be realized within a single regime without violating extension-invariant identity or reintroducing hidden discriminators.

The regimes introduced below are persistence-and-admissibility profiles induced when the requirements of Section 4 must be satisfied simultaneously under the neutrality and stability constraints fixed in Section 3.

7.1 Goal and Proof Strategy

The goal of this section is to prove a numeric lower bound on the number of referential regimes required to satisfy R1–R6 simultaneously under the optimization criterion fixed in Section 3.

The argument has three steps.

First, we extract from R2–R6 (subject to R1) a set of structurally distinct persistence-and-admissibility profiles that must be realizable in any compliant substrate. These profiles correspond to candidate referential regimes forced by the requirements.

Second, we state a collapse admissibility criterion: a proposed identification of two regimes is admissible only if it preserves extension-invariant identity and admissible relations without introducing interpretive flags, roles, or context-dependent predicates at the extension layer.

Third, we show that any collapse among the forced profiles violates the admissibility criterion through one of three failure modes: identity instability, category error, or hidden regime reintroduction. It follows that the substrate must realize at least as many distinct regimes as there are forced profiles, establishing the lower bound.

7.2 Induced referential Regimes

We now determine which referential regimes are forced by R2–R6 when satisfied simultaneously under R1’s requirement of extension-invariant identity and admissible relations.

An referential regime is characterized by (i) persistence conditions and (ii) the class of relations admissible without reclassification.

When the requirements are satisfied simultaneously under R1, their persistence and admissibility constraints give rise to distinct candidate referential-

regime profiles. We extract these profiles in turn.

7.2.1 Profile induced by R2: Obligation-bearing persistence

R2 requires representation of entities that bear obligations independently of particular occurrences. Such entities must persist across compliance, violation, delegation, and discharge. Their admissible relations include normative attribution without altering identity. This induces an initial distinct persistence-and-admissibility profile: obligation-bearing.

7.2.2 Profile induced by R3: Normative-structure persistence

R3 requires representation of normative or regulatory structures independently of their execution. Such entities persist across enactment, amendment, interpretation, and repeal. Their admissible relations include structural modification without collapsing into occurrences. This induces a second distinct profile: normative-structure.

7.2.3 Profile induced by R4: Time-indexed occurrence persistence

R4 requires representation of occurrences individuated by temporal realization and provenance. Such entities do not persist through modification but are fixed by their time-indexed realization. This induces a third distinct persistence profile incompatible with enduring entities: time-indexed occurrence.

7.2.4 Profile induced by R5: Applicability-context persistence

R5 requires representation of scopes or applicability contexts as enduring entities. Such entities admit nesting, overlap, and conditional scoping without collapsing into either authority or bearer entities. This induces a fourth

structurally distinct profile: applicability-context.

7.2.5 Profile induced by R6: Descriptive-record persistence

R6 requires representation of descriptive indicators without causal or normative commitment. Such entities admit measurement and comparison relations without implying execution or authority. This induces a fifth distinct profile: descriptive-record.

7.2.6 Enduring Non-Normative Referents

Satisfaction of R2–R5 presupposes stable reference to enduring entities that are neither obligation-bearing nor authority-bearing, yet may be regulated, inspected, or otherwise affected by occurrences.

Such entities persist independently of normative attribution and independently of time-indexed realization.

Their persistence conditions are those of enduring referents whose identity is not altered by normative attribution, structural amendment, or temporal realization.

Their admissible relations include participation in occurrences as patients or objects of action, as well as measurement and description relations, without implying normative authority or responsibility.

Profiles are individuated by the conjunction of persistence conditions and admissible-relation classes. Two candidate regimes are distinct whenever either their persistence criteria or their admissible-relation profiles differ.

For example, although ENR and REC both admit measurement and description relations, their persistence conditions and participation roles differ, yielding distinct identity-and-admissibility profiles.

This profile is not independently required by any single requirement. However,

satisfying R2–R5 simultaneously presupposes stable reference to enduring entities that may participate in occurrences or be subject to regulation or inspection without thereby bearing obligations or grounding authority.

Such entities must admit participation and measurement relations without admitting normative attribution relations.

This admissibility profile is incompatible with that of obligation-bearing entities, whose structural relations include normative attribution. If they were identified with obligation-bearing entities, normative attribution relations would become structurally admissible for all enduring referents, or implementations would require a discriminator to block such attributions. Either outcome violates the admissibility criterion introduced below.

Accordingly, the representational capacities required by R2–R5 induce a distinct persistence-and-admissibility profile for enduring non-normative referents.

This yields a ninth distinct persistence-and-admissibility profile: enduring non-normative referents.

7.2.7 Resulting Candidate Partition

The induced profiles are pairwise non-equivalent under R1’s stability constraint. They differ either in persistence conditions or in admissible-relation classes.

Accordingly, any substrate satisfying R1–R6 must realize at least nine distinct referential regimes. For reference, we fix the following abbreviations for these regimes:

1. **OBL: Obligation-bearing entity:** an enduring referent that may bear obligations or responsibilities and remain identifiable as a party across time.
2. **NOR: Normative structure:** an enduring referent that grounds

obligations or permissions without being tied to any single occurrence.

3. **OCC: Time-indexed occurrence:** an entity individuated by its temporal realization and provenance.
4. **CTX: Applicability context:** a referent that scopes where, to whom, or under what conditions authority applies, and that may be nested or overlapping.
5. **REC: Descriptive record:** a referent that reports measured or derived properties without asserting causal or normative conclusions.
6. **ENR: Enduring non-normative referent:** an enduring entity that may be regulated, inspected, or otherwise affected by occurrences, but does not itself ground obligations or bear normative responsibility.

7.3 Collapse Admissibility Criterion

To establish a lower bound, we consider whether any two of the induced referential regimes can be collapsed into a single regime without violating the constraints of Section 3.

A collapse of two regimes is admissible only if the merged representation realizes a single stable reference and persistence profile and supports a unified class of admissible relations without introducing interpretive flags, role predicates, or contextual discriminators at the extension layer, which would reintroduce regime distinctions contrary to the neutrality and extension-stability constraints of Section 3.

Under R1, identity and admissibility must be invariant across incompatible extensions. Accordingly, a collapse is inadmissible if it requires any of the following:

- **Identity instability:** the merged regime requires incompatible persistence criteria, so that identity judgments vary across extensions.

- **Category error:** the merged regime structurally admits relations inappropriate to some of its instances, requiring interpretive blocking rules.
- **Hidden regime reintroduction:** the original distinction is recovered through flags, roles, or contextual predicates, thereby reintroducing the regime at the extension layer.

Any collapse exhibiting one of these failure modes violates either extension-invariant identity or the neutrality criterion excluding hidden discriminators.

7.4 Pairwise Collapse Analysis

There are finitely many unordered pairs among the induced regimes. We consider each class of collapse and show that every such identification violates the admissibility criterion by instantiating one of the three failure modes defined above.

7.4.1 Category error: Obligation-bearing and Enduring non-normative referent

Suppose an obligation-bearing entity and an enduring non-normative referent are collapsed into a single regime.

The merged regime must admit a unified class of admissible relations. Obligation-bearing entities admit normative attribution relations such as *hasObligation* or *isResponsibleFor* without reclassification.

If those relations are structurally admissible for all instances of the merged regime, then every enduring referent becomes eligible for normative attribution in principle, embedding a normative commitment at the substrate level.

If, instead, normative attribution is restricted to only some instances, implementations must introduce a discriminator such as *isObligationBearer*

to block inappropriate attributions. The original distinction is thereby reintroduced at the extension layer.

In either case, the collapse is inadmissible.

7.4.2 Identity instability: Obligation-bearing and Normative structure

Suppose an obligation-bearing entity and a normative structure are collapsed into a single regime.

Obligation-bearing entities persist across compliance, violation, and delegation. Normative structures persist across enactment, amendment, and repeal.

These persistence criteria are non-equivalent. No single identity condition can preserve both without making persistence judgments vary across extensions.

The collapse therefore violates extension-invariant identity under R1.

7.4.3 Category error: Normative structure and Applicability context

Suppose a normative structure and an applicability context are collapsed into a single regime.

Normative structures admit structural modification relations such as enactment, amendment, delegation, and repeal. Applicability contexts instead admit scoping relations such as nesting, overlap, and conditional refinement.

If the two are identified, the merged regime must admit both modification-type and scoping-type relations as structurally admissible. However, these relation classes serve fundamentally different roles. Amendment is a structural modification that preserves the identity of a normative structure across change. Whereas nesting and overlap are relations among scopes that

organize applicability conditions without modifying normative content.

No single admissible-relation profile can unify these relation classes. Either the relations are admitted uniformly, producing category error for some instances, or their admissibility must be restricted contextually, reintroducing the distinction through a hidden discriminator.

If all relations are admitted uniformly, entities that are merely applicability scopes become amendable normative structures, producing category error.

If relation admissibility is restricted by a predicate such as *isNormative* or *isScope*, the original distinction is reintroduced as a hidden regime at the extension layer.

The collapse is therefore inadmissible.

7.4.4 Hidden regime reintroduction: Normative structure and Descriptive record

Suppose a normative structure and a descriptive record are collapsed into a single regime.

Normative structures ground obligations and admit normative evaluation, enforcement, and modification. Descriptive records report measured or derived properties without asserting causal or normative commitments.

If both are identified, the merged regime must admit a unified admissible-relation profile. If normative evaluation relations are structurally admissible for all instances of the merged regime, then purely descriptive records become eligible for normative grounding, embedding normative commitments at the substrate level.

If, instead, normative evaluation is restricted to a subset of instances, implementations must introduce a discriminator such as *isNormative* or *isDescriptive* to distinguish grounding entities from descriptive ones. Such a discriminator reintroduces the original distinction at the extension layer,

violating the admissibility criterion and the neutrality constraint.

The collapse is therefore inadmissible.

7.4.5 Summary of representative collapse failures

Collapsing time-indexed occurrences with enduring non-normative referents produces identity instability. Occurrence identity is fixed by temporal realization, whereas enduring referents persist independently of time-indexed realization. Accordingly, no unified persistence criterion can satisfy both.

Table 1 summarizes representative pairwise collapses and the failure mode each induces under the admissibility criterion defined in Section 7.3.

Table 1: Representative collapse failures under neutrality and stability constraints

Collapsed regimes	Failure mode	Reason for inadmissibility
OBL / ENR	Category error	Normative attribution relations appropriate to obligation-bearing entities would become structurally admissible for all enduring referents, misclassifying non-normative entities as potential obligation-bearers.
OBL / NOR	Identity instability	Persistence as a responsible party and persistence as a normative structure require incompatible identity criteria across amendment and restructuring.
NOR / CTX	Category error	Modification relations (enactment, amendment, repeal) conflict structurally with scoping relations (nesting, overlap, conditional refinement); no unified admissible-relation profile can serve both without misclassifying instances of one type as the other.
NOR / REC	Hidden regime reintroduction	Normative grounding and purely descriptive reporting admit incompatible admissibility profiles; implementations reintroduce <i>isNormative</i> or <i>isDescriptive</i> flags to recover the distinction.
OCC / ENR	Identity instability	Time-indexed individuation criteria conflict with extension-invariant persistence required for enduring entities.

These three failure modes are exhaustive relative to the admissibility criterion.

Since the admissibility criterion requires exactly three conditions to hold jointly (extension-invariant identity, uniform admissibility of relations for the merged regime, and neutrality with respect to hidden discriminators) any collapse can violate admissibility only by failing one of these conditions. Identity instability corresponds to violation of the first, category error to violation of the second, and hidden regime reintroduction to violation of the third.

The remaining pairwise collapses instantiate the same three failure modes: identity instability, category error, or hidden regime reintroduction. Under the admissibility criterion of Section 7.3, no collapse eliminates a regime without violating neutrality or stable reference (see Appendix 17 for the complete enumeration of all fifteen pairs).

Lemma 7.1 (Lower bound on referential regimes). *Under the assumptions of Section 3, any ontology satisfying requirements R1–R6 and the collapse admissibility criterion of Section 7.3 must realize at least nine distinct referential regimes.*

Proof sketch. Section 7.2 shows that satisfying R1–R6 under extension-invariant identity induces nine pairwise non-equivalent persistence-and-admissibility profiles, each corresponding to a candidate referential regime.

Any ontology realizing fewer than nine regimes must identify at least two of these profiles within a single regime. By the collapse admissibility criterion (Section 7.3), every such identification produces either identity instability, category error, or hidden regime reintroduction, as demonstrated by exhaustive pairwise analysis (Section 7.4 and Appendix 17).

Each of these outcomes violates the neutrality or stability constraints fixed in Section 3.

Therefore no ontology with fewer than nine regimes can satisfy R1–R6 simultaneously. $\square$

7.5 Scope of the Necessity Result

The necessity result is conditional on the assumptions of Section 3. It does not claim that fewer regimes are impossible in general, but that they are incompatible with the optimization objective fixed in Section 3. Ontologies optimized for local agreement, negotiated semantics, or expressive flexibility may legitimately employ fewer explicit distinctions by relying on hidden or interpretive mechanisms.

Within the stated constraints, the result establishes a lower bound: any neutral substrate capable of supporting accountability under persistent disagreement must realize at least nine distinct referential regimes.

8 A Six-Family Construction Before Refinement

8.1 The Initial Six Referential Regimes

This section exhibits a construction realizing the six initial referential regimes derived in Section 7. No additional regimes are introduced, and none of the induced regimes are merged or weakened.

Each regime is implemented with the identity conditions and admissible-relation profile specified in the derivation. The construction fixes these regimes as pairwise disjoint, with extension-invariant persistence criteria, and without interpretive flags or role-based discriminators.

We now specify a minimal set of typed relations that connects these regimes while preserving neutrality and stable reference.

8.2 Admissible Relations

The construction introduces the following typed relations. Together these relations realize the admissibility profiles required to satisfy R2–R6. Each

requirement is realized by at least one listed relation. The coverage mapping is made explicit in Section 8.3.

- authority grounding and delegation:

$$\textit{enacts}, \textit{issues} : \text{OBL} \rightarrow \text{NOR}$$

- participation and party status:

$$\textit{party-to} : \text{OBL} \rightarrow \text{NOR}$$

$$\textit{involves} : \text{OCC} \rightarrow \text{OBL}$$

$$\textit{has-patient} : \text{OCC} \rightarrow \text{ENR}$$

- execution under authority:

$$\textit{occurs-under} : \text{OCC} \rightarrow \text{NOR}$$

- applicability:

$$\textit{applies-in} : \text{NOR} \rightarrow \text{CTX}$$

$$\textit{refines} : \text{CTX} \rightarrow \text{CTX}$$

$$\textit{overlaps} : \text{CTX} \leftrightarrow \text{CTX}$$

- description:

$$\textit{measures} : \text{REC} \rightarrow (\text{OBL} \mid \text{ENR} \mid \text{CTX})$$

$$\textit{observes} : \text{REC} \rightarrow \text{OCC}$$

Relations in this construction are structural and descriptive. They assert reference, participation, scoping, or provenance, but do not encode causal mechanism or normative validity.

Authority grounding and delegation records that an obligation-bearing entity enacts or issues a normative structure, without asserting validity or enforceability.

Participation relations record that an occurrence is associated with obligation-bearing entities via *involves* and with enduring non-normative referents via *has-patient*.

The *measures* relation associates a descriptive record with an enduring refer-

ent, while *observes* anchors a descriptive record to a time-indexed occurrence. Neither relation asserts accuracy, causal relevance, or normative significance.

Occurrences serve as temporal and provenance anchors for participation and descriptive relations. Descriptive records target enduring referents whose identity persists across incompatible extensions.

This relation set suffices to satisfy R1–R6 without introducing additional identity regimes, role-based typing, or extension-layer discriminators.

Temporal and provenance information attach only to OCC and REC via typed attributes or admissible relations. These annotations do not introduce new identity criteria or additional identity-bearing categories.

8.3 Satisfaction of Structural Requirements

The relation set defined above satisfies R1–R6 by the following coverage mapping:

- **R1 (Stable identity):** Each regime fixes an extension-invariant referential criterion. No relation in the construction propagates persistence criteria across regime boundaries.
- **R2 (Obligation-bearing capacity):** *enacts*, *issues*, *party-to*, and *involves* realize normative attribution for OBL without reclassification.
- **R3 (Normative reference without execution):** *enacts*, *issues*, and *applies-in* ground NOR independently of any OCC instance.
- **R4 (Time-indexed occurrence):** *involves*, *has-patient*, *occurs-under*, and *observes* anchor OCC to its participants and governing normative structure without altering their identity.
- **R5 (Applicability and scope):** *applies-in* scopes NOR within CTX; *refines* realizes conditional nesting; *overlaps* realizes non-hierarchical partial scope without requiring containment.
- **R6 (Descriptive indicators):** *measures* associates REC with enduring referents (OBL, ENR, CTX) without asserting causal or normative

conclusions.

No requirement introduces a relation that propagates persistence criteria or normative commitments outside the regime for which it is specified. No additional identity regimes or extension-layer discriminators are required.

Proof. By construction, each of R2–R6 is realized by at least one typed relation in the construction, and all regimes satisfy R1. The regimes are pairwise disjoint, and the typed relation set connects them only through descriptive, participation, and scoping relations whose admissibility conditions do not propagate persistence criteria across regime boundaries. and the construction introduces no interpretive flags, role-based discriminators, or additional identity regimes. Hence the construction satisfies R1–R6. $\square$

The preceding results should be read in two stages. First, R1–R6 force a coarse six-family structure: obligation-bearing entities, normative structures, time-indexed occurrences, applicability contexts, descriptive records, and enduring non-normative referents. Second, transformation-driven refinement shows that three of those families cannot remain single regime profiles: ENR splits into ENR-L and ENR-I, CTX splits into CTX-E and CTX-S, and NOR splits into NOR-C and NOR-S.

The operative lower-bound result is therefore Theorem 10.4: under the stated assumptions, a compliant substrate must realize at least nine pairwise non-collapsing regime profiles. This is a current lower bound, not a tight bound, not an upper bound, and not a completeness claim.

9 Regime Refinement

This section derives required refinements of regime profiles under transformation-driven identity constraints. Refinement occurs when no single transformation-classification assignment is consistent across all admissible cases within a provisional regime.

9.1 Refinement Principle

Definition 9.1 (Refinement condition). A regime profile P requires refinement if there exists a transformation family $f \in \mathbb{F}$ such that no single classification

$$\text{Class}_P(f) \in \{\text{IGN}, \text{PRS}, \text{BRK}\}$$

is consistent with all admissible instances assigned to P .

Definition 9.2 (Forced refinement). A refinement is forced if failure to refine yields violation of stable reference.

Refinement separates a provisional regime into distinct profiles that realize incompatible transformation-classification patterns.

9.2 ENR Refinement

Proposition 9.3 (ENR split under branching). *The regime of enduring non-normative referents requires refinement under branching transformations BF.*

Proof. Under BF, an identity carrier may produce multiple continuations. In some admissible cases, identity must be preserved across branches; in others, branching produces distinct identities.

No single classification of BF as PRS or BRK is consistent across all such cases. Therefore, refinement is required. $\square$

Definition 9.4 (ENR refinement). The ENR regime splits into:

- ENR-L: identity bound to locus (branching breaks identity),
- ENR-I: identity bound to instrument (branching preserves identity across instances).

9.3 CTX Refinement

Proposition 9.5 (CTX split under decomposition). *The regime of applicability contexts requires refinement under aggregation/decomposition transformations AD.*

Proof. Under AD, a context may be decomposed into subcontexts. In some admissible cases, identity is preserved under decomposition; in others, decomposition alters applicability conditions.

No single classification of AD as PRS or BRK is consistent across all admissible cases. Therefore, refinement is required. $\square$

Definition 9.6 (CTX refinement). The CTX regime splits into:

- CTX-E: extension-sensitive contexts,
- CTX-S: structure-sensitive contexts.

9.4 NOR Refinement

Proposition 9.7 (NOR split under refinement). *The regime of normative structures requires refinement under refinement transformations RF.*

Proof. Under RF, a structure may be refined into more detailed form. In some admissible cases, identity is preserved under refinement; in others, refinement alters structural composition in a way that changes identity.

No single classification of RF as PRS or BRK is consistent across all admissible cases. Therefore, refinement is required. $\square$

Definition 9.8 (NOR refinement). The NOR regime splits into:

- NOR-C: composition-sensitive structures,
- NOR-S: substitution-sensitive structures.

9.5 Summary of Refinement

Theorem 9.9 (Refinement result). *The provisional regime set requires refinement into distinct profiles for ENR, CTX, and NOR under transformation families BF, AD, and RF respectively.*

Proof. Propositions 9.3, 9.5, and 9.7 establish that no single profile can realize consistent classification for the corresponding transformation families. Therefore, refinement into distinct profiles is required. $\square$

Refinement is driven entirely by transformation-classification inconsistency. No additional ontological distinctions are introduced.

9.6 Refinement of Enduring Non-Normative Referents

This subsection shows that the class of enduring non-normative referents does not define a single regime profile under the transformation basis, and must be refined.

9.6.1 Initial ENR Profile

Definition 9.10 (Provisional ENR profile). A provisional ENR profile is a regime profile whose identity carrier is an enduring non-normative referent, i.e., an enduring entity that does not itself ground obligations or normative structure.

Under this provisional formulation, identity is intended to track persistence of an enduring referent independently of normative or causal interpretation.

9.6.2 Transformation Behavior

We examine the classification of transformations under the provisional ENR profile.

- RE: IGN
- AN: IGN
- RF: PRS
- AD: PRS
- RC: IGN
- RA: N/A
- SU: BRK
- BF: PRS or BRK
- PV: IGN
- SE: PRS

The classification of branching is indeterminate under the provisional ENR profile: it is identity-preserving for locus-bound referents but identity-breaking for instrument-bound referents. This indeterminacy is the source of split pressure.

9.6.3 Split Pressure

Proposition 9.11 (ENR split pressure). *The provisional ENR profile is under split pressure with respect to the branching transformation BF.*

Proof of Proposition 9.11. Let $x \in \mathcal{R}$ be a locus-bound representation and let τ be a branching transformation producing continuations over the same locus. Since the locus persists through branching, $\text{Class}_P(\text{BF}) = \text{PRS}$ is required, i.e., branching must be identity-preserving.

Let $x' \in \mathcal{R}$ be an instrument-bound representation and let τ' be a branching transformation producing two non-identical artifact continuations. At most

one continuation can be the same artifact as x' , so $\text{Class}_P(\text{BF}) = \text{BRK}$ is required, i.e., branching must be identity-breaking.

These requirements are contradictory. No single value of $\text{Class}_P(\text{BF})$ is consistent with both classes of cases. Therefore the provisional ENR profile is under split pressure with respect to BF. $\square$

[Informally] After branching, the locus remains the same but the artifact does not.

9.6.4 Separated Profiles

We therefore define two distinct regime profiles.

Definition 9.12 (ENR-L profile). The locus-bound profile ENR-L is defined by:

- carrier: enduring locus,
- identity basis: persistence of the same locus,
- transformation classification:
 - RE: IGN
 - AN: IGN
 - RF: PRS
 - AD: PRS
 - RC: IGN
 - RA: N/A
 - SU: BRK
 - BF: PRS
 - PV: IGN
 - SE: PRS

Definition 9.13 (ENR-I profile). The instrument-bound profile ENR-I is defined by:

- carrier: enduring artifact or instrument,
- identity basis: persistence of the same artifact,
- transformation classification:
 - RE: IGN
 - AN: IGN
 - RF: PRS
 - AD: PRS
 - RC: IGN
 - RA: N/A
 - SU: BRK
 - BF: BRK
 - PV: IGN
 - SE: PRS

9.6.5 Non-Collapse

Proposition 9.14 (ENR-L and ENR-I do not collapse). *The profiles ENR-L and ENR-I induce distinct equivalence relations on $\mathcal{R}$.*

Proof of Proposition 9.14. Let $x \in \mathcal{R}$ be an instrument representation. Let τ be a branching transformation producing a distinct artifact continuation $y := \tau(x)$ over the same locus as x .

Under ENR-L:

$$\text{Class}_{\text{ENR-L}}(\text{BF}) = \text{PRS},$$

so τ is identity-preserving and $x \sim_{\text{ENR-L}} y$.

Under ENR-I:

$$\text{Class}_{\text{ENR-I}}(\text{BF}) = \text{BRK},$$

so τ is identity-breaking and $x \not\sim_{\text{ENR-I}} y$.

Therefore $\sim_{\text{ENR-L}} \neq \sim_{\text{ENR-I}}$. □

[Informally] After branching, two representations can remain identical as loci even when they are no longer the same artifact. Thus the profiles define different equivalence classes.

9.6.6 Result

Theorem 9.15 (ENR refinement). *The class of enduring non-normative referents must be refined into exactly two distinct regime profiles under the branching transformation BF:*

$$\text{ENR-L} \quad \text{and} \quad \text{ENR-I}.$$

[Informally] Two representations can refer to the same locus even when the artifacts differ, but cannot refer to the same artifact once branching produces distinct objects.

Exactly two profiles suffice: BF admits only PRS or BRK, and each is realized by one of the two profiles.

9.7 Refinement of Applicability Contexts

This subsection shows that applicability contexts do not define a single regime profile under the transformation basis, and must be refined.

9.7.1 Initial CTX Profile

Definition 9.16 (Provisional CTX profile). A provisional CTX profile is a regime profile whose identity carrier is an applicability scope or condition, i.e., a structure that determines where, to whom, or under what conditions authority applies.

Under this provisional formulation, identity is intended to track persistence of applicability independently of internal structure.

9.7.2 Transformation Behavior

We examine the classification of transformations under the provisional CTX profile.

- RE: IGN
- AN: IGN
- RF: PRS or BRK
- AD: PRS or BRK
- RC: PRS or BRK
- RA: IGN or BRK
- SU: BRK
- BF: PRS or BRK
- PV: IGN
- SE: not applicable ($\text{Appl}_P(\text{SE}) = 0$; normative structures do not undergo state evolution).

The classification of decomposition is indeterminate under the provisional CTX profile: it is identity-preserving when applicability extension is unchanged, but identity-breaking when internal structure is changed. This indeterminacy is the source of split pressure.

9.7.3 Split Pressure

Proposition 9.17 (CTX split pressure). *The provisional CTX profile is under split pressure with respect to the decomposition transformation AD.*

Proof of Proposition 9.17. Let $x \in \mathcal{R}$ be a context representation whose identity is determined by applicability extension: x represents a jurisdiction

defined as the set of cases falling under a given authority, independently of how that jurisdiction is internally organized. Let τ be a decomposition transformation producing subcontexts whose union covers the same set of cases as x , and let $y := \tau(x)$. Since the applicability extension of y equals that of x , identity-preservation is required: $\text{Class}_P(\text{AD}) = \text{PRS}$.

Let $x' \in \mathcal{R}$ be a context representation whose identity is determined by internal structure: x' represents a partition of cases into named subregions, where the partition itself constitutes the context. Let τ' be a decomposition transformation that subdivides x' into finer subregions, producing $y' := \tau'(x')$ with different internal organization. Since the structure of y' differs from that of x' , identity-breaking is required: $\text{Class}_P(\text{AD}) = \text{BRK}$.

Both x and x' are admissible instances of the provisional CTX profile, as both are applicability scopes independent of normative or causal content. The requirements PRS and BRK are contradictory. Therefore the provisional CTX profile is under split pressure with respect to AD. $\square$

[Informally] After decomposition, the applicability can remain the same even when the structure does not.

9.7.4 Separated Profiles

We therefore define two distinct regime profiles.

Definition 9.18 (CTX-E profile). The extensional applicability profile CTX-E is defined by:

- carrier: applicability scope or condition,
- identity basis: persistence of the same applicability extension,
- transformation classification:
 - RE: IGN
 - AN: IGN
 - RF: PRS

- AD: PRS
- RC: PRS (re-contextualization preserves applicability extension by definition of CTX-E)
- RA: IGN (reassignment does not alter applicability extension)
- SU: BRK
- BF: PRS
- PV: IGN
- SE: not applicable ($\text{Appl}_P(\text{SE}) = 0$; normative structures do not undergo state evolution).

Definition 9.19 (CTX-S profile). The structured applicability profile CTX-S is defined by:

- carrier: applicability structure,
- identity basis: persistence of the same structure,
- transformation classification:
 - RE: IGN
 - AN: IGN
 - RF: BRK
 - AD: BRK
 - RC: BRK (re-contextualization alters applicability structure)
 - RA: BRK (reassignment alters the structural bearer of applicability)
 - SU: BRK
 - BF: BRK
 - PV: IGN
 - SE: not applicable ($\text{Appl}_P(\text{SE}) = 0$; normative structures do not undergo state evolution).

The classifications of RC and RA are fully determined by the identity basis of each profile. Under CTX-E, both are PRS or IGN because neither alters applicability extension. Under CTX-S, both are BRK because each alters

the structural configuration that constitutes identity.

This does not induce further split pressure. In both profiles, re-contextualization or reassignment is identity-preserving if and only if the defining identity basis is unchanged: applicability extension for CTX-E, and applicability structure for CTX-S.

Thus the variation in classification reflects dependence on whether the underlying invariant is preserved, rather than the presence of distinct regime profiles.

9.7.5 Non-Collapse

Proposition 9.20 (CTX-E and CTX-S do not collapse). *The profiles CTX-E and CTX-S induce distinct equivalence relations on $\mathcal{R}$.*

Proof of Proposition 9.20. Let $x \in \mathcal{R}$ be a context representation. Let τ be a decomposition transformation producing subcontexts whose union has the same applicability extension as x , and let $y := \tau(x)$.

Under CTX-E:

$$\text{Class}_{\text{CTX-E}}(\text{AD}) = \text{PRS},$$

so τ is identity-preserving and $x \sim_{\text{CTX-E}} y$.

Under CTX-S:

$$\text{Class}_{\text{CTX-S}}(\text{AD}) = \text{BRK},$$

so τ is identity-breaking and $x \not\sim_{\text{CTX-S}} y$.

Therefore $\sim_{\text{CTX-E}} \neq \sim_{\text{CTX-S}}$. □

[Informally] After decomposition, two representations can have the same applicability even when their structure differs.

9.7.6 Result

Theorem 9.21 (CTX refinement). *Applicability contexts must be refined into at least two distinct regime profiles:*

$$\text{CTX-E} \quad \text{and} \quad \text{CTX-S}.$$

[Informally] Two representations can apply to the same cases even when their internal structure differs, but cannot be the same structure once decomposition changes that organization.

9.8 Refinement of Normative Structures

This section shows that normative structures do not define a single regime profile under the transformation basis, and must be refined.

9.8.1 Initial NOR Profile

Definition 9.22 (Provisional NOR profile). A provisional NOR profile is a regime profile whose identity carrier is a normative structure, i.e., an enduring referent that grounds obligations or permissions independently of any particular occurrence.

Under this provisional formulation, identity is intended to track persistence of a normative structure independently of representation or organization.

9.8.2 Transformation Behavior

We examine the classification of transformations under the provisional NOR profile.

- RE: IGN

- AN: IGN
- RF: PRS or BRK
- AD: PRS or BRK
- RC: IGN
- RA: IGN
- SU: BRK
- BF: BRK
- PV: IGN
- SE: not applicable ($\text{Appl}_P(\text{SE}) = 0$; normative structures do not undergo state evolution).

The classification of refinement is indeterminate under the provisional NOR profile: it is identity-preserving when normative content is unchanged, but identity-breaking when structural organization changes. This indeterminacy is the source of split pressure.

9.8.3 Split Pressure

Proposition 9.23 (NOR split pressure). *The provisional NOR profile is under split pressure with respect to the refinement transformation RF.*

Proof of Proposition 9.23. Let $x \in \mathcal{R}$ be a normative structure and let τ be a refinement transformation producing a representation $y := \tau(x)$ that preserves the same normative content.

If identity is determined by normative content, then $\text{Class}_P(\text{RF}) = \text{PRS}$ is required, i.e., refinement must be identity-preserving.

If identity is determined by structural organization, then $\text{Class}_P(\text{RF}) = \text{BRK}$ is required, i.e., refinement must be identity-breaking.

These requirements are contradictory. No single value of $\text{Class}_P(\text{RF})$ is consistent with both cases. Therefore the provisional NOR profile is under

split pressure with respect to RF. □

[Informally] After refinement, the same normative content can remain even when the structure does not.

9.8.4 Separated Profiles

We therefore define two distinct regime profiles.

Definition 9.24 (NOR-C profile). The content-preserving normative profile NOR-C is defined by:

- carrier: normative structure,
- identity basis: persistence of the same normative content,
- transformation classification:
 - RE: IGN
 - AN: IGN
 - RF: PRS
 - AD: PRS
 - RC: IGN
 - RA: IGN
 - SU: BRK
 - BF: BRK
 - PV: IGN
 - SE: not applicable ($\text{Appl}_P(\text{SE}) = 0$; normative structures do not undergo state evolution).

Definition 9.25 (NOR-S profile). The structural normative profile NOR-S is defined by:

- carrier: normative structure,
- identity basis: persistence of the same structure,
- transformation classification:

- RE: IGN
- AN: IGN
- RF: BRK
- AD: BRK
- RC: IGN
- RA: IGN
- SU: BRK
- BF: BRK
- PV: IGN
- SE: not applicable ($\text{Appl}_P(\text{SE}) = 0$; normative structures do not undergo state evolution).

Both NOR-C and NOR-S share the same carrier (normative structure). The refinement is by identity basis rather than carrier type: normative content versus structural organization.

9.8.5 Non-Collapse

Proposition 9.26 (NOR-C and NOR-S do not collapse). *The profiles NOR-C and NOR-S induce distinct equivalence relations on $\mathcal{R}$.*

Proof of Proposition 9.26. Let $x \in \mathcal{R}$ be a normative structure and let τ be a refinement transformation producing a representation $y := \tau(x)$ with the same normative content.

Under NOR-C:

$$\text{Class}_{\text{NOR-C}}(\text{RF}) = \text{PRS},$$

so τ is identity-preserving and $x \sim_{\text{NOR-C}} y$.

Under NOR-S:

$$\text{Class}_{\text{NOR-S}}(\text{RF}) = \text{BRK},$$

so τ is identity-breaking and $x \not\sim_{\text{NOR-S}} y$.

Therefore $\sim_{\text{NOR-C}} \neq \sim_{\text{NOR-S}}$. □

[Informally] After refinement, two representations can express the same normative content even when their structure differs.

9.8.6 Result

Theorem 9.27 (NOR refinement). *Normative structures must be refined into at least two distinct regime profiles:*

$$\text{NOR-C} \quad \text{and} \quad \text{NOR-S}.$$

[Informally] Two representations can encode the same normative content even when organized differently, but cannot be the same structure once that organization changes.

10 Nine Regimes of Referential Identity

This paper has addressed a structural question that arises when ontologies are required to function as neutral substrates under persistent disagreement. Under explicit constraints of neutrality, stability under incompatible extension, and the exclusion of hidden structural discriminations, we have shown that a lower bound exists on the individuation structure required to support accountability. Ontologies with fewer than nine distinct referential regimes cannot satisfy the stated neutrality and stability requirements under the assumptions fixed in Section 3.

This section states the current lower-bound result after refinement. The result should be read as a lower bound within the stated formal setting: the derived nine profiles are pairwise non-collapsing under the adopted transformation basis. It does not establish that no additional regimes are possible, nor does it claim completeness for arbitrary ontological systems. The sufficiency

construction shows that this boundary is achievable: a substrate realizing at least nine referential regimes satisfies all stated requirements without hidden or excess structure. The result establishes that certain distinctions are unavoidable under the stated assumptions, yielding a current lower bound on the referential-regime structure required to support accountability under persistent disagreement.

The analysis shows that accountability under persistent disagreement requires a cross-cutting stability constraint governing identity and persistence across all represented entities, together with explicit representation of obligation-bearing entities, normative structures, time-indexed occurrences, applicability contexts, and descriptive records, each governed by distinct identity conditions.

The result is conditional rather than prescriptive. It does not claim that the identified structure is optimal for all applications, nor does it assess the adequacy of alternative modeling strategies for other objectives. It establishes only that, under a particular and explicitly stated optimization objective, certain distinctions are structurally unavoidable.

This work clarifies the design space for neutral substrates under persistent disagreement. Under the stated constraints, it shows that the lower bound on explicit ontological structure is determined by structural requirements rather than by modeling preference. Future work may examine how alternative optimization objectives yield different structural bounds, or analyze how the nine-regime construction behaves when stability or neutrality assumptions are systematically weakened. Such analyses may help clarify how different interoperability objectives generate different minimal ontological structures.

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The author reviewed all suggestions and takes full responsibility for the content of this work.

11 Regime Structure and Neutral Representation

This section consolidates the regime profiles established by the preceding derivations and states the current lower bound on the number of distinct referential regimes required.

11.1 Previously Established Regimes

The following regime profiles were defined in Section 5 and are not refined by the transformations analyzed in Sections 9.6–9.8:

- OBL: obligation-bearing entities,
- OCC: time-indexed occurrences,
- REC: descriptive records.

These profiles do not exhibit split pressure under the transformation families considered thus far. Their classifications remain internally consistent with respect to identity preservation.

11.2 Refined Regimes

The preceding sections establish the following refinements:

- ENR $\Rightarrow$ ENR-L, ENR-I,
- CTX $\Rightarrow$ CTX-E, CTX-S,
- NOR $\Rightarrow$ NOR-C, NOR-S.

Each refinement is forced by split pressure under a specific transformation:

- ENR splits under branching (BF),
- CTX splits under decomposition (AD),
- NOR splits under refinement (RF).

11.3 Derived Regime Set

This subsection fixes the regime profiles derived in Sections 9.6, 9.7, and 9.8. These profiles form the basis for the lower bound and pairwise non-collapse results.

Definition 11.1 (Derived regime set). The derived regime set consists of the following nine regime profiles:

{OBL, OCC, REC, ENR-L, ENR-I, CTX-E, CTX-S, NOR-C, NOR-S}.

Each regime profile is determined by its identity carrier and its classification behavior over the fixed transformation basis.

- OBL: obligation-bearing entity profile.
- OCC: time-indexed occurrence profile.
- REC: descriptive record profile.
- ENR-L: enduring referent (locus-bound).
- ENR-I: enduring referent (instrument-bound).
- CTX-E: applicability context (extension-sensitive).
- CTX-S: applicability context (structure-sensitive).
- NOR-C: normative structure (composition-sensitive).
- NOR-S: normative structure (substitution-sensitive).

These profiles are defined entirely by their identity and persistence behavior under the transformation basis. No additional ontological categories are introduced.

Convention 11.2. $\mathcal{S}$ denotes an ontology known to be admissible. $\mathcal{O}$ denotes an arbitrary ontology whose admissibility is under consideration in a given proof or argument.

11.4 Regime-Typed Multigraph Embedding

Definition 11.3 (Regime-typed directed multigraph). Let $\mathcal{S}$ be an admissible ontology with derived regime set $\mathcal{P}$. The regime-typed directed multigraph $G(\mathcal{S}) = (V, E, \lambda, \rho)$ is defined by:

- $V = \bigsqcup_{P \in \mathcal{P}} \mathcal{E}_P$, the disjoint union of equivalence classes over all regime profiles,
- $E \subseteq V \times \mathcal{A} \times V$, where $\mathcal{A}$ is the admissible relation vocabulary,
- $\lambda : V \rightarrow \mathcal{P}$ assigns each vertex its regime type,
- $\rho : E \rightarrow \mathcal{A}$ labels each edge with its admissible relation.

Theorem 11.4 (Faithful embedding). *The map $\iota : \mathcal{S} \rightarrow G(\mathcal{S})$ sending each identity carrier x to its equivalence class $[x]_P$ under its unique regime profile P (Lemma 11.8) is injective up to regime-typed isomorphism and preserves admissible structure.*

Proof. Injectivity follows from regime determinacy (Lemma 11.8): distinct identity behaviors map to distinct equivalence classes. Admissible structure is preserved because edges in $G(\mathcal{S})$ are drawn exclusively from the admissible relation vocabulary $\mathcal{A}$, and no causal or normative relations are introduced. Regime-typed isomorphism is the appropriate equivalence because identity is defined at the level of equivalence classes, not individual representations. $\square$

This embedding is the construction asserted in the abstract and consequences sections. Path families over $G(\mathcal{S})$ constitute the representational basis for

structural comparison and transformation. Full development of path grammar and expressive adequacy is given in Section 6.

11.5 Lower Bound

Theorem 11.5 (Current lower bound). *Any substrate satisfying the requirements of Sections 5–9.8 must realize at least nine pairwise non-collapsing regime profiles.*

Proof. Sections 9.6, 9.7, and 9.8 establish non-collapse for the pairs (ENR-L, ENR-I), (CTX-E, CTX-S), and (NOR-C, NOR-S).

These refinements establish nine distinct profiles: ENR-L, ENR-I, CTX-E, CTX-S, NOR-C, and NOR-S.

Together with the previously established profiles OBL, OCC, and REC, this yields nine candidate regime profiles.

Pairwise non-collapse for this regime set is established in Section 11.6. Therefore the substrate must realize at least nine pairwise non-collapsing regime profiles. $\square$

11.6 Pairwise Non-Collapse Matrices

To organize the pairwise non-collapse argument, we record the status of each unordered pair of derived regime profiles in matrix form. The first matrix gives the coarse status of each pair: entries marked THM are already separated by prior split results, while entries marked GEN are separated by a generic cross-family criterion. The next two matrices refine the generic entries by indicating the kind of separation used.

Generic separation criteria. For distinct profiles P, Q , a matrix entry marked GEN indicates that P and Q differ in at least one of the following

regime-defining respects: (i) carrier of identity, (ii) identity basis, (iii) applicability classification over the fixed transformation basis, or (iv) persistence behavior. Any such difference that changes the induced equivalence relation yields non-collapse.

Coarse pairwise non-collapse matrix.

	OBL	OCC	REC	ENR-L	ENR-I	CTX-E	CTX-S	NOR-C	NOR-S
OBL	–	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>
OCC	<i>GEN</i>	–	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>
REC	<i>GEN</i>	<i>GEN</i>	–	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>
ENR-L	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	–	THM	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>
ENR-I	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	THM	–	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>
CTX-E	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	–	THM	<i>GEN</i>	<i>GEN</i>
CTX-S	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	THM	–	<i>GEN</i>	<i>GEN</i>
NOR-C	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	–	THM
NOR-S	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	<i>GEN</i>	THM	–

For the refined matrices, THM indicates separation established by an earlier split theorem. GEN-CAR indicates separation established by difference in identity carrier. All pairs that share an identity carrier are separated by the split theorems, and no additional separation by identity basis is required for the derived regime set. Therefore, the matrix contains no GEN-BAS entries.

Refined pairwise non-collapse matrix, Part I.

	OBL	OCC	REC	ENR-L	ENR-I
OBL	–	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
OCC	GEN-CAR	–	GEN-CAR	GEN-CAR	GEN-CAR
REC	GEN-CAR	GEN-CAR	–	GEN-CAR	GEN-CAR
ENR-L	GEN-CAR	GEN-CAR	GEN-CAR	–	THM
ENR-I	GEN-CAR	GEN-CAR	GEN-CAR	THM	–
CTX-E	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
CTX-S	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
NOR-C	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
NOR-S	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR

Refined pairwise non-collapse matrix, Part II.

	CTX-E	CTX-S	NOR-C	NOR-S
OBL	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
OCC	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
REC	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
ENR-L	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
ENR-I	GEN-CAR	GEN-CAR	GEN-CAR	GEN-CAR
CTX-E	–	THM	GEN-CAR	GEN-CAR
CTX-S	THM	–	GEN-CAR	GEN-CAR
NOR-C	GEN-CAR	GEN-CAR	–	THM
NOR-S	GEN-CAR	GEN-CAR	THM	–

11.7 Exhaustiveness of Regime Profiles

Lemma 11.6 (Profile realizability for admissible ontologies). *Let $x \in \mathcal{R}$ be an identity carrier belonging to an admissible ontology. Then there exists a regime profile P in the derived regime set such that the identity of x is given by the equivalence class $[x]_P$.*

Proof. Because x belongs to an admissible ontology, its identity and persistence conditions are invariant under admissible extension (Section 5). By Definition 5.14, identity under a regime profile is determined by the equivalence relation induced by transformations classified as PRS.

Since the transformation basis $\mathbb{F}$ is fixed (Section 5.3), the identity behavior of x under admissible transformations is fully determined by its classification over $\mathbb{F}$ together with its identity basis. By the remark following Definition 5.14, no additional structure contributes to identity.

If this transformation-classification pattern is already realized by one of the previously established regime profiles, then the claim follows immediately. If not, then by Definition 5.23, failure of realization under a single profile constitutes split pressure. Such failure must arise from conflicting identity requirements under at least one transformation family in $\mathbb{F}$.

Sections 9.6, 9.7, and 9.8 analyze all transformation families that generate such conflicts for identity carriers within the structural scope of this analysis, and establish the corresponding forced refinements.

Therefore, every admissible identity behavior considered in this paper is realized by some regime profile in the derived regime set, and the identity of x is given by the corresponding equivalence class $[x]_P$. $\square$

11.8 Uniqueness of Regime Realization

Lemma 11.7 (Non-overlap of regime realization). *No admissible identity carrier in the present analysis has identity behavior realized by more than one regime profile in the derived regime set.*

Proof. Suppose x is realized by two regime profiles P and Q in the derived regime set. Then the identity of x is given by both $[x]_P$ and $[x]_Q$, so the induced identity relations coincide on the identity behavior of x .

If P and Q were distinct realizations of the same identity behavior, then their

induced equivalence relations would coincide, i.e., $\sim_P = \sim_Q$ on the relevant domain. By Definition 5.18, this implies that P and Q collapse.

However, pairwise non-collapse for all profiles in the derived regime set is established in Section 11.6. Therefore no two profiles in the derived regime set collapse, and no identity carrier is realized by more than one profile.

Therefore, no admissible identity carrier is realized by more than one regime profile. $\square$

11.9 Regime Determinacy

Lemma 11.8 (Regime determinacy). *For every identity carrier x in an admissible ontology, there exists exactly one regime profile P such that the identity of x is given by the equivalence class $[x]_P$.*

Proof. Existence follows from Lemma 11.6. Uniqueness follows from Lemma 11.7. $\square$

11.10 Representation Theorem

Theorem 11.9 (Neutral representation theorem). *Let $\mathcal{S}$ be an admissible ontology over the representation space $\mathcal{R}$ with transformation basis $\mathbb{F}$. Assume identity carriers in $\mathcal{S}$ are among the structural forms considered in Sections 9.6–9.8.*

Then for every identity carrier $x \in \mathcal{S}$, there exists exactly one regime profile P in the derived regime set such that the identity of x is given by the equivalence class $[x]_P$.

Proof. Because $\mathcal{S}$ is admissible, identity and persistence conditions for each identity carrier are invariant under admissible extensions (Section 5).

By Lemma 11.6, the identity behavior of each $x \in \mathcal{S}$ is realized by some regime profile P in the derived regime set.

By Lemma 11.7, no identity carrier is realized by more than one regime profile.

Therefore, each identity carrier in $\mathcal{S}$ admits representation by exactly one regime profile, with identity given by the equivalence class $[x]_P$. $\square$

The theorem establishes representational adequacy of the derived regime set for admissible ontologies within the formal setting of this paper. It does not claim that arbitrary ontological systems admit such a representation.

12 Consequences

The result establishes a lower bound of nine on the number of distinct referential regime profiles required for admissible substrates. This has several structural consequences.

First, individuation is constrained by invariance: identity and persistence must be fixed independently of interpretation. Systems that rely on contextual or role-based reclassification for identity violate admissibility.

Second, relations operate over entities without determining their identity. This enforces a separation between reference and interpretation, ensuring that participation, measurement, and scoping relations do not alter identity conditions.

Third, regime distinctions are forced by transformation-induced split pressure. When a transformation must be identity-preserving under one identity basis and identity-breaking under another, a single regime is insufficient and refinement is required. The derived splits of ENR, CTX, and NOR are instances of this phenomenon.

12.1 Illustrative Examples

The following examples illustrate the three transformation-induced splits derived in Sections 9.6–9.8. They serve only to anchor interpretation and do not contribute to the derivation.

ENR (branching)

Consider a physical location and an artifact. Under branching, multiple continuations may refer to the same location, but distinct artifacts cannot remain identical. This forces a distinction between locus-bound and instrument-bound regimes.

CTX (decomposition)

Consider a jurisdiction defined as a union of subregions. Decomposition into subcontexts can preserve the same applicability while altering internal structure. This forces a distinction between extension-based and structure-based contexts.

NOR (refinement)

Consider a normative rule expressed at different levels of detail. Refinement can preserve the same normative content (what the rule requires or permits) while altering its structural organization. A content-tracking identity basis classifies such refinement as identity-preserving; a structure-tracking basis classifies it as identity-breaking. This forces a distinction between content-sensitive (NOR-C) and structure-sensitive (NOR-S) regimes.

12.2 Consequences for Representation

The embedding establishes that all admissible structure is representable as regime-typed path families over admissible relations. Accordingly:

- structural comparison reduces to comparison of path families,

- admissibility reduces to preservation of path structure under transformation,
- identity is evaluated at the level of equivalence classes, not individual representations.

12.3 Consequences for Interpretation

Interpretive content is not represented at the substrate level. All interpretive assignments arise in extensions over REC-anchored paths.

Thus:

- identical path structure may admit multiple incompatible interpretations,
- interpretive disagreement does not affect identity or persistence,
- causal and normative content are external to admissible structure.

12.4 Consequences for Comparison

Two systems are structurally comparable if and only if their path families are comparable under regime-typed isomorphism.

Therefore:

- comparison is invariant under admissible re-expression and refinement,
- differences in interpretation do not affect structural comparison,
- non-comparability arises only from differences in regime structure or admissible relations.

12.5 Consequences for Intervention

Interventions correspond to admissible transformations of path structure.

An intervention is minimal if it alters the admissibility or classification of a transformation family without introducing non-admissible structure.

Accordingly:

- modifying NOR alters admissibility of pathways,
- modifying CTX alters applicability domains,
- modifying ENR alters capacity to instantiate pathways,
- modifying REC alters observability of path structure,
- modifying OCC alters the time-indexed occurrences through which path transitions are realized.

Interventions propagate along admissible paths and are evaluated by their effect on path families, not by interpretive outcomes.

12.6 Consequences for Structural Analysis

Structural analysis reduces to the study of path families under admissible transformations.

In particular:

- detection of distortion corresponds to identification of anomalous or asymmetric path structure,
- explanation corresponds to identification of transformation families responsible for such structure,
- correction corresponds to minimal modification of regime-level constraints affecting those transformation families.

No appeal to causal or normative primitives is required for detection, comparison, or correction at the structural level.

13 Interpretation of the Result

This section clarifies how the lower- and upper-bound results should be understood, and how they should not be understood. The aim is not to defend the result against alternative modeling goals, but to situate it relative to the explicit optimization criterion fixed earlier in the paper.

13.1 Combined Characterization and Scope

Taken together, the necessity and sufficiency results establish a conditional optimality claim. Under the assumptions of Section 3, any ontology intended to function as a neutral substrate for accountability under persistent disagreement must realize at least nine distinct referential regimes. Conversely, a construction with a minimal set of nine such regimes suffices to meet all stated representational requirements without introducing hidden structure or interpretive commitments.

The result is a conditional lower bound: it establishes that no representation with fewer than nine regimes can satisfy the stated requirements under the adopted optimization criterion. It does not merely show adequacy or inadequacy under particular conditions; it establishes that at least nine regimes are necessary under the stated optimization objective.

13.2 Minimality Relative to Stability

The result should not be read as a claim about simplicity in an intuitive or engineering sense. Ontologies with fewer explicit entity kinds may be easier to describe or extend within a coordinated interpretive community. However, when such reductions are achieved by encoding stability-critical distinctions as roles, flags, or contextual predicates, total structural complexity is not reduced. Instead, it is displaced into extension layers whose interpretation cannot be held fixed under disagreement.

In this paper, minimality is defined relative to stability under persistent disagreement. *A representation is minimal only if removing explicit structure does not require reintroducing the same distinctions implicitly.* From this perspective, an ontology with fewer explicit kinds but greater interpretive burden is not simpler, but less explicit.

13.3 Why Hidden Regimes Are Excluded

Hidden regimes are often effective where interpretation is shared, enforced, or low-cost to renegotiate. They support flexibility and expressive economy within a single governance framework.

The present analysis excludes such regimes not because they are incorrect, but because they are incompatible with the stated objective. When disagreement is persistent and uncoordinated, distinctions that are not fixed at the substrate level cannot be relied upon to preserve identity or admissible relations. Stability then depends on interpretive convergence, which the substrate is explicitly designed not to assume.

13.4 Relation to Other Modeling Strategies

Many modeling frameworks adopt optimization objectives different from those adopted here, including expressive richness, domain coverage, implementation convenience, explanatory power, or alignment with existing legal, scientific, or institutional vocabularies. Such frameworks may introduce additional categories, contextual typing, role predicates, or domain-specific commitments precisely in order to support richer forms of representation and use.

The result presented here neither subsumes nor contradicts those strategies. It identifies a specific point in the design space: the referential-regime structure required, under the stated assumptions, for a neutral substrate operating under persistent interpretive disagreement. Frameworks optimized for other

goals may occupy different points in that space, including designs that add further distinctions or accept forms of contextual classification excluded by the neutrality and stable-reference constraints used in this paper.

The following comparison is illustrative rather than evaluative. Table 2 shows common representation strategies for these regimes in BFO 2.0 and DOLCE and highlights where regime-level distinctions required for neutrality are not structurally enforced.

The comparison is structural rather than taxonomic; it identifies where regime-level distinctions required for neutrality are left to implementation rather than enforced at the type level.

The table identifies the structural under-constraint that would arise in each case. These under-constraints are not problems in their intended use. They are points where neutral substrate implementations built on those frameworks might need to introduce additional modeling commitments that the present substrate treats as regime-level structure.

The nine regimes are not derived from gaps in existing frameworks. Rather, independently motivated requirements reveal where those frameworks leave regime-level distinctions to the implementer.

Table 3 summarizes the systematic under-constraints that arise in both frameworks relative to the neutrality and stability requirements.

13.5 Conditional Scope of the Claim

The lower-bound result is explicitly conditional. If any assumption of Section 3 is rejected, such as neutrality, stability under incompatible extension, or the exclusion of hidden regimes, the optimization criterion is invalidated and the bound no longer applies. We do not claim that these assumptions are universally correct; we claim only that the result follows when they are adopted.

Table 2: Illustrative representation strategies for the nine regimes in BFO 2.0 and DOLCE, with structural under-constraints under neutrality and stability requirements.

Regime	BFO 2.0	DOLCE
ENR (enduring non-normative referent)	Independent Continuant (material object). Clean for material entities; institutional and fiat entities require additional modeling choices not enforced by the framework.	Physical or Social Object (endurant). Social object is first-class, providing a cleaner fit for institutional referents than BFO.
OBL (obligation-bearing entity)	Specifically Dependent Continuant (Role or Disposition) inhering in an Independent Continuant. Obligation-bearing is not a first-class category; it is realized via a role-bearer pair, splitting what the present substrate treats as a single regime.	Quality or property of a Social Object. Obligation-bearing capacity is not a structural first-class distinction; the OBL/NOR boundary remains implementation-dependent.
NOR (non-normative structure)	Generically Dependent Continuant (information artifact, normative document). The substrate-level separation between authority-as-referent (NOR) and authority-as-executed (OCC) is a modeling choice, not structurally enforced.	Social Object (normative structure). Better motivated than BFO, but authority and execution can both be represented as social objects or perdurants without a regime-level firewall between NOR and OCC.
OCC (time-indexed occurrence)	Process or Process Boundary (occurrent). Closest mapping in both frameworks; BFO occurrents are well-individuated temporally.	Perdurant (Event or Process). DOLCE distinguishes achievements from activities, which is richer than required but not incompatible.
CTX (applicability context)	Site (immaterial Independent Continuant) for spatial scope only. Non-spatial applicability contexts (normative jurisdiction, conditional scope) have no stable BFO home and are represented as GDC, conflating CTX with NOR.	Abstract Region or Feature. DOLCE regions are spatiotemporal; normative applicability collapses into Social Object (conflating CTX with NOR) or Quality (conflating CTX with REC).
REC (descriptive indicators without causal commitment)	Generically Dependent Continuant (data item, measurement information entity). Both normative structures (NOR) and descriptive indicators (REC) map to GDC; the stability-critical distinction between them is not explicitly represented at the BFO type level.	Quality (with associated quale or region). DOLCE’s quality theory is the strongest match; the without-causal-commitment constraint is a modeling discipline, however, not structurally enforced.

If stability is weakened to allow context-dependent regime membership (e.g., role- or viewpoint-relative identity/admissibility), some regimes may be representable as role- or parameter-dependent refinements of fewer base regimes, but doing so shifts structural burden into coordination, versioning, or exchange mechanisms whose admissibility conditions are not fixed at the substrate level and are outside the scope of this analysis; the lower bound therefore does not apply under such weakenings.

Within the stated scope, however, the result is unavoidable. Under the stated assumptions, any attempt to reduce the number of regimes would require sacrificing stability, reintroducing hidden distinctions, or weakening the neutrality constraint. The nine-regime structure thus emerges not as a design preference, but as a structural consequence of the problem as posed.

Table 3: Systematic under-constraints in BFO 2.0 and DOLCE relative to neutrality and stability requirements.

Under-constraint	BFO 2.0	DOLCE
NOR/REC conflated	Both normative structures and descriptive indicators map to GDC; the stability-critical distinction is not explicitly represented at the type level and left to the implementer.	Both map to Social Object or Quality depending on implementation; the distinction is a modeling discipline, not structurally enforced.
CTX has no structural home	Non-spatial applicability contexts are represented as GDC, effectively conflating CTX with NOR.	Normative applicability collapses into Social Object (conflating CTX with NOR) or Quality (conflating CTX with REC).
OBL/NOR boundary implementation-dependent	The separation between obligation-bearer and normative structure is mediated by the inheres-in relation, not enforced by regime membership; unstable under delegated authority.	The same boundary is mediated by participation relations; regime-level enforcement is absent.

Implementations that resolve these under-constraints silently introduce hidden regimes in the sense defined in Section 7.

The present analysis does not model state transitions, versioning mechanisms, or evolution protocols. Such structures may be required for practical interoperability, but they operate downstream of the regime-level identity constraints established here.

13.6 Implications

The primary implication of this result is diagnostic rather than prescriptive. It provides a principled way to evaluate claims that an ontology supports neutral interoperability under disagreement. If such a framework relies on implicit roles or context-dependent typing for stability-critical distinctions, the present result explains why those claims cannot be sustained simultaneously under the stated constraints.

Conversely, for designers who share the stated optimization objective, the result identifies which distinctions must be made explicit, and which may safely be deferred to extension layers, in order to preserve neutrality and stable reference across incompatible interpretations. The nine-regime structure thus serves both as a lower bound and as a design template for accountability substrates in contested domains.

The resulting regime structure can be understood as an axiomatization pattern for substrates intended to support divergent specialization: identity and admissibility constraints are fixed at the substrate level so that subsequent causal or normative extensions may diverge without retroactive reinterpretation. Here, "axiomatization" is used in a structural sense: it fixes identity conditions and admissible relations as substrate invariants, while leaving causal and normative commitments to be introduced by extensions.

13.7 Grounding the Construction in the Running Cases

The nine-regime construction is not an abstract partition. Each case introduced in Section 1 exercises the full structure and illustrates what stable

reference under persistent disagreement concretely requires.

13.7.1 Multi-jurisdiction regulatory reporting.

A corporate entity is represented as an obligation-bearing referent (OBL) that persists across jurisdictional differences in how corporate personhood or subsidiary liability is defined. The reporting statute under which it operates is represented as an authority-bearing structure (NOR) that persists across amendments, contested interpretations, and delayed enforcement. Each filing, inspection, or transaction is represented as a time-indexed occurrence (OCC) individuated by when and where it happened, independently of how any jurisdiction evaluates it. The jurisdiction itself is represented as an applicability context (CTX) that scopes which obligations apply to which entities without collapsing into either the statute or the filer. Performance and compliance indicators are represented as descriptive records (REC) that remain comparable across jurisdictions without carrying causal or normative commitments. Any entity that is regulated but does not itself bear obligations, such as a regulated substance or a monitored facility, is represented as an acted-upon referent (ENR). The nine-regime construction allows two jurisdictions to refer to the same corporation, the same statute, and the same filing while maintaining incompatible evaluations of compliance, without either forcing interpretive convergence or losing the shared referential ground required for audit comparison.

13.7.2 Legislative history tracking.

A statute is represented as an authority-bearing structure (NOR) that remains the same normative referent across the legislative acts that enacted it (OCC), the amendments that modified it (OCC), and the administrative decisions that applied it (OCC). The legislature and the agencies that implement the statute are represented as obligation-bearing referents (OBL) whose identity does not depend on any particular enactment event. Jurisdictions and

conditional scopes such as the geographic or subject-matter boundaries of the statute’s application, are represented as applicability contexts (CTX) that may be nested, overlapping, or contested without destabilizing reference to the statute itself. Records of legislative proceedings, judicial interpretations, and compliance histories are represented as descriptive indicators (REC) that characterize outcomes without asserting which interpretation of the statute is correct. The construction allows legal historians, policy analysts, and regulators working under incompatible interpretive frameworks to refer to the same statutory authority and the same sequence of enactment events while disagreeing entirely on what the statute means or which application was legitimate.

13.7.3 Cross-institutional provenance.

A research or administrative record is represented as an acted-upon referent (ENR) whose identity and persistence conditions are invariant under the classificatory schemes of any receiving institution. The institution that produced the record and the institutions that subsequently hold or analyze it are represented as obligation-bearing referents (OBL) that remain identifiable as custodians across transfers, mergers, and re-organizations. Policies governing data use, retention, or sharing are represented as authority-bearing structures (NOR) that persist independently of whether they have been enforced or contested in any particular transfer. Each transfer, annotation event, or re-analysis is represented as a time-indexed occurrence (OCC) individuated by its provenance, independently of how institutions classify the action. Institutional scope conditions, such as which data a policy governs or which entities fall under a data-sharing agreement, are represented as applicability contexts (CTX). Derived indicators, quality metrics, and outcome summaries attached to the record are represented as descriptive records (REC) that remain comparable across institutions whose causal and normative frameworks differ. The construction preserves the provenance chain as a stable referential structure even when no two institutions agree on how the record

should be classified, evaluated, or interpreted.

14 Related Work

This paper is situated at the intersection of formal ontology, knowledge representation, and the study of classification under disagreement. Rather than surveying ontology engineering broadly, this section situates the present result relative to prior work on classification under disagreement, identity and roles in formal ontology, and upper-ontology design.

14.1 Classification, Disagreement, and Neutrality

A substantial body of work has shown that classification systems embed institutional, social, and political commitments rather than functioning as neutral descriptive devices (Bowker and Star 1999; Haslanger 2012). In scientific and policy contexts, disagreement is often persistent rather than transient, reflecting incompatible evidentiary, methodological, or evaluative frameworks rather than resolvable empirical uncertainty (Longino 1990).

These observations motivate the need for representations that remain usable without assuming interpretive convergence. The present work adopts this motivation but addresses a distinct structural question: which referential-regime distinctions become unavoidable when disagreement is treated as persistent and causal or normative commitments are externalized from the substrate rather than settled within it. The answer given here is conditional on the neutrality, stable-reference, and no-hidden-regime assumptions stated above: under those assumptions, the analysis yields a current lower bound on the referential-regime structure required for accountability under persistent disagreement.

14.2 Identity, Roles, and Ontological Methodology

Formal ontology has long emphasized the role of identity conditions and persistence criteria in category formation. The OntoClean methodology (Guarino and Welty 2002) clarifies why roles are anti-rigid and unsuitable as identity-defining categories (Guarino 1998), and subsequent work on social and institutional entities has further developed these distinctions (Masolo et al. 2003).

This paper builds on these insights but applies them at the substrate level. The present analysis operates at a different level than role-based modeling, deriving substrate-level referential regimes from neutrality requirements rather than from domain classifications.

14.3 Upper Ontologies

Upper ontologies such as BFO (Smith and Ceusters 2015), DOLCE (Gangemi et al. 2002), and UFO (Guizzardi 2005) provide rich accounts of dependence, endurants and perdurants, and social objects. They differ substantially in commitments and optimization objectives, and are not designed primarily as neutral substrates for interoperability under persistent disagreement.

The present work does not compare expressivity or advocate alignment with any particular upper ontology. Instead, it derives lower-bound constraints that apply independently of such frameworks, conditional on the neutrality and stability requirements adopted here. The regime structure derived here can be situated with respect to two widely-used frameworks to clarify where under-constraints arise not as defects in those frameworks but as points where implementations built on them would need to introduce additional commitments to satisfy the neutrality and stability requirements assumed here.

BFO (Smith and Ceusters 2015) organizes entities under the continuant/occurrent distinction: continuants persist through time while occurrents unfold across

it (Arp et al. 2015). OCC (time-indexed occurrences) maps cleanly onto BFO occurrents. ENR through NOR map onto independent and generically dependent continuants, but BFO does not enforce the regime-level separation between them: obligation-bearing entities (OBL), authority-bearing structures (NOR), and descriptive indicators (REC) are all candidates for Generically Dependent Continuant, leaving their stability-critical distinctions to the implementer. CTX (applicability and scope) has no first-class BFO home; non-spatial applicability contexts such as normative jurisdiction or conditional scope are represented as GDC, effectively conflating CTX with NOR under a BFO-based implementation. Under persistent disagreement, these conflations require hidden discriminators to recover regime-level distinctions, introducing implicit structure of the kind prohibited by the neutrality constraint.

DOLCE (Gangemi et al. 2002) introduces social objects as first-class entities and provides a richer treatment of institutional and normative structure than BFO (Masolo et al. 2003). This makes DOLCE a better fit for OBL and NOR as distinct categories. However, the NOR/REC conflation persists: both normative structures and descriptive indicators are candidates for social object or quality, and the distinction between authority-as-referent and descriptive-record-as-referent is not enforced at the type level. DOLCE’s region-based account of scope is spatiotemporal, leaving normative applicability contexts (CTX) without a structural home distinct from NOR or REC, and the OBL/NOR boundary (between obligation-bearer and normative structure) is mediated by participation relations rather than enforced by regime membership (Masolo et al. 2003). As with BFO, implementations resolving these under-constraints under persistent disagreement silently introduce hidden regimes.

These observations do not constitute objections to BFO or DOLCE as general-purpose ontological frameworks. They locate a specific point where the optimization objectives of those frameworks (expressiveness, metaphysical coverage, alignment with scientific practice) diverge from the optimization objective assumed here, and where that divergence has structural consequences

for neutral interoperability under persistent disagreement.

When constrained to enforce conservative extension over identity and admissibility conditions, such frameworks can realize the regime structure derived here. The contribution of this paper is to identify the structural constraints required when neutrality under persistent disagreement is taken as the primary objective.

The present result is therefore not a competing upper ontology, but a constraint on any ontology intended to function as a neutral, stability-preserving substrate under persistent disagreement.

14.4 Relation to Common Logic and Ontology Exchange

The neutral substrate and derived regimes can be expressed in ISO/IEC 24707 Common Logic (ISO/IEC 2018) as a conservative theory, supporting interpreter-independent ontology exchange without reliance on description-logic restrictions or fixed signatures. Here, conservative means that extensions add structure without altering the consequences derivable about substrate-level terms (Hodges 1997), ensuring that incompatible interpretive frameworks do not retroactively change the meaning of shared references. This enables compatibility with ontology exchange workflows widely used in applied ontology (Kutz et al. 2010) and digital twin infrastructures (Grieves and Vickers 2016), where multiple interpretive layers must interoperate over stable referents.

14.5 Causality, Explanation, and Infrastructure

Separating descriptive structure from causal or explanatory claims is a well-established principle in causal analysis (Pearl 2009). Similarly, research on large-scale information infrastructures emphasizes the importance of long-term stability and interpretive restraint across institutional boundaries (Edwards 2011).

These perspectives align with the design neutrality constraints assumed in this paper. The ontology analyzed here is not an explanatory theory, but an infrastructural substrate intended to support explanation, evaluation, and policy analysis without embedding them.

14.5.1 Relation to FAIR Data Principles

The FAIR principles promote data reuse across communities and over time, but do not specify the ontological constraints required for interoperability under persistent disagreement. In practice, reuse is often pursued through negotiated vocabularies or shared interpretations, which presuppose forms of convergence that may not be available in legal, political, or analytic settings.

The results of this paper clarify a structural precondition for FAIR-aligned reuse in pluralistic or institutionally heterogeneous contexts: stable reference must be supported independently of causal or normative agreement. The referential regimes derived here therefore constrain the ontological substrates upon which FAIR-compliant systems implicitly rely when reuse occurs under persistent disagreement.

This work is situated at the intersection of ontology, knowledge representation, and data modeling, with a focus on identity, persistence, and reference under disagreement.

14.6 Identity and Persistence

Philosophical and formal treatments of identity have long distinguished criteria for persistence across time and change. Approaches based on endurantism and perdurantism provide different accounts of how entities persist, but typically assume a fixed interpretive framework within which identity conditions are evaluated.

In contrast, this work considers identity conditions that must remain stable under persistent interpretive disagreement. The requirement that identity

be invariant under admissible transformations introduces constraints that are not captured by standard accounts of persistence.

14.7 Ontology and Knowledge Representation

In ontology engineering and knowledge representation, identity is often tied to classification, typing, or role assignment, as in description logics and related frameworks. Such systems typically allow identity conditions to depend on context, interpretation, or modeling choices.

The present work departs from these approaches by requiring that identity conditions be independent of causal and normative commitments at the substrate level. Rather than constructing representations, it derives structural constraints that any substrate must satisfy to support neutral reference across interpretive frameworks.

14.8 Data Modeling and Provenance

Data models and provenance frameworks track entities across transformations, including versioning, aggregation, and derivation. Standards such as the W3C PROV model explicitly represent derivation and lineage relations, while temporal and versioned data models capture change over time.

These systems provide mechanisms for tracking entities under transformation, but may allow identity to depend on representation, derivation history, or application-specific semantics.

This work abstracts from such domain-specific considerations and instead identifies transformation-invariant identity regimes required for stable reference. The transformation families considered here provide a uniform basis for analyzing identity behavior across domains.

14.9 Position of This Work

Unlike prior work that assumes a fixed interpretive setting, this paper derives referential regimes from invariance under admissible transformations in a neutral substrate. The result is a necessary lower bound on the number of regimes required to maintain stable reference under persistent interpretive disagreement. No claim is made of completeness or minimality.

15 Limits and Boundary Conditions

This section specifies the boundary of applicability of the result and identifies classes of systems excluded by the admissibility constraints.

15.1 Scope of the Result

The result applies only to admissible substrates (Section 5), namely those in which identity conditions, persistence conditions, and admissible relations are invariant under admissible extension and independent of interpretive commitment.

In particular, the result concerns neutral substrates under persistent interpretive disagreement, and establishes only a necessary lower bound on the number of regime profiles required to maintain stable identity under such conditions.

The result is structural: it characterizes identity and persistence behavior in terms of invariance under a fixed transformation basis, and does not depend on any domain-specific interpretation.

15.2 Excluded Systems

The following classes of systems are excluded from the domain of the result:

- **Interpretively dependent identity:** Systems in which the identity condition of an entity depends on interpretive commitments realized in extensions. Such systems violate identity invariance and do not admit a neutral substrate.
- **Relation-determined identity:** Systems in which relations introduce or alter identity conditions for their relata. Relations defined over pre-established equivalence classes are admissible; relations that constitute or modify those classes are not.
- **Extension-variant identity or relations:** Systems in which identity conditions or admissible relations vary under extension of the represented domain. Such systems violate extension stability.
- **Embedded interpretive commitments:** Systems that encode interpretive content at the substrate level whose validity depends on extension-specific assumptions. Such systems violate interpretive non-commitment.
- **Non-admissible transformation semantics:** Systems requiring transformation behavior not representable within the fixed transformation basis $\mathbb{F}$. Such systems fall outside the structural scope of the analysis.

15.3 Non-Goals

This paper does not:

- establish an upper bound or claim completeness of the regime set;
- construct or analyze interpretive extensions;
- provide domain-specific modeling guidance;
- assign causal or normative interpretation to structural representations;
- establish completeness or minimality of the transformation basis $\mathbb{F}$, or determine whether additional transformation families would force further regime splits.

15.4 Boundary of Representation

The representation preserves identity, regime typing, and admissible relations under transformation. It does not preserve interpretive content.

Accordingly:

- causal and normative interpretations arise only in extensions,
- interpretive disagreement does not affect identity or persistence,
- structural equivalence does not imply interpretive equivalence.

The limits identified in this section are consequences of the neutrality constraint and the restriction to admissible transformation behavior. They are not limitations of a particular representation technique, but intrinsic to the structural setting considered.

16 Conclusion

17 Closing Synthesis

This paper establishes that admissible substrate systems admit a neutral, regime-typed representation in which structure is realized as families of admissible paths.

The construction is:

- structurally sufficient for admissible multi-stage systems,
- faithful up to regime-typed isomorphism,
- neutral with respect to causal and normative interpretation.

Interpretation is deferred to extensions over REC-anchored path families. Comparison reduces to path comparison. Intervention reduces to admissible path transformation.

The representation does not encode causal or normative force. Where such commitments are needed, they are introduced by extension layers rather than by the substrate itself.

The result establishes that at least nine pairwise non-collapsing regime profiles are required under the stated assumptions, and that admissible structure within the finite-system setting can be represented as regime-typed path families. This provides a representational basis for subsequent analysis over path structures subject to the admissibility constraints adopted here.

This paper derives structural constraints on identity and persistence for neutral substrates under persistent interpretive disagreement. Within this setting, identity is determined by invariance under admissible transformations rather than by causal explanation or normative evaluation.

We show that transformation-induced split pressure requires refinement of certain regime profiles. Enduring non-normative referents split under branching, applicability contexts split under decomposition, and normative structures split under refinement, each producing distinct profiles required for stable identity.

These results establish a necessary lower bound on the number of referential regimes required for such substrates. At least nine distinct regime profiles are required. Completeness or minimality of this set is not established.

The result is structural and conditional. It characterizes constraints that any substrate must satisfy in order to support stable reference under admissible transformations, without embedding causal or normative commitments.

Statements and Declarations

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Author Contributions

The author is the sole contributor to this work and is responsible for all aspects of the research, authorship, and publication.

Use of AI-Assisted Tools

AI-assisted tools were used for editing, formatting, and consistency checking. The author reviewed all suggestions and is solely responsible for the content.

Declaration of Conflicting Interest

The author declares no potential conflicts of interest with respect to the research, authorship, and publication of this work.

Appendix A. Complete Pairwise Collapse Enumeration

All $\binom{6}{2} = 15$ unordered pairs among the nine regimes are enumerated below and classified by primary failure mode under the admissibility criterion of Section 7.3. Each pair appears exactly once in the enumeration.

Identity Instability

nine pairs cannot be collapsed without violating extension-invariant identity.

- OBL/NOR (incompatible persistence criteria: obligation-bearing entities persist across delegation and compliance, whereas normative structures persist across enactment, amendment, and repeal);
- OBL/OCC (incompatible persistence criteria: obligation-bearing entities are enduring referents, whereas occurrences are individuated by temporal realization and do not persist through modification);
- OCC/ENR (incompatible persistence criteria: occurrences are time-indexed realizations, whereas enduring referents persist independently of temporal realization);
- OCC/NOR (incompatible persistence criteria: occurrences are individuated by realization events, whereas normative structures persist across enactment, amendment, and repeal);
- OCC/CTX (incompatible persistence criteria: occurrences are time-indexed realizations, whereas applicability contexts persist as enduring scoping referents);
- OCC/REC (incompatible persistence criteria: occurrences are individuated by temporal realization, whereas descriptive records persist as enduring referents whose identity is not determined by any single realization event).

Category Error

nine pairs cannot be collapsed without producing a category error under the neutrality constraint.

- OBL/ENR (normative attribution relations appropriate to obligation-bearing entities would become structurally admissible for all enduring referents, misclassifying non-normative entities as potential obligation-bearers);
- OBL/CTX (obligation-bearing entities admit normative attribution relations, whereas applicability contexts admit scoping relations, yielding incompatible admissible-relation classes);
- NOR/CTX (normative structures admit modification relations such as enactment and amendment, whereas applicability contexts admit scoping relations such as nesting and refinement);
- NOR/ENR (normative grounding relations appropriate to normative structures would misclassify purely enduring referents as sources of authority);
- CTX/ENR (scoping relations appropriate to applicability contexts would misclassify ordinary enduring referents as scope-defining entities);
- CTX/REC (scoping relations appropriate to applicability contexts would misclassify descriptive records as scope-defining referents).

Hidden Regime Reintroduction

Three pairs cannot be collapsed without reintroducing the original distinction through hidden discriminators.

- NOR/REC (normative grounding relations would require an *isNormative* discriminator to prevent descriptive records from functioning as normative authorities);
- OBL/REC (normative attribution relations would require a discrimina-

tor such as *isObligationBearer* to prevent descriptive records from being treated as obligation-bearing entities, reintroducing the distinction at the extension layer);

- ENR/REC (measurement targets and descriptive records would require an *isDescriptive* flag to distinguish descriptive records from the entities they describe).

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